

University of Engineering and Technology, Lahore

Narowal Campus

**Piezoelectric-Based Smart Helmet
Accident Detection System**

A Thesis Submitted in Partial Fulfilment of the Requirements for the
Degree of

Bachelor of Science in Electrical Engineering

By

Farheen Fayyaz (2022-EE-604)

Waseem Ahmed (2022-EE-619)

Raja Sana Ullah (2022-EE-620)

Session: 2022–2026

Supervisor

Engr. Amna Javed

June, 2026

Declaration

We, Farheen fayyaz (2022-EE-604), Waseem Ahmed (2022-EE-619), and Raja Sana Ullah (2022-EE-620), hereby state that my BS thesis titled “Piezoelectric-Based Smart Helmet Accident Detection System” is my own work and has not been submitted previously by me for taking any degree from University of Engineering and Technology, Lahore or anywhere else in the world.

At any time if our statement is found to be incorrect even after our graduation, the university has the right to withdraw our BS degree.

June, 2026

Farheen Fayyaz (2022-EE-604)

Waseem Ahmed (2022-EE-619)

Raja Sana Ullah (2022-EE-620)

Plagiarism Undertaking

We, Farheen Fayyaz (2022-EE-604), Waseem Ahmed (2022-EE-619), and Raja Sana Ullah (2022-EE-620), solemnly declare that research work presented in the thesis titled “Piezoelectric-Based Smart Helmet Accident Detection System” is solely our research work with no significant contribution from any other person. Small contribution/help wherever taken has been duly acknowledged and that complete thesis has been written by ourselves.

We understand the zero tolerance policy of the HEC and University of Engineering and Technology, Lahore towards plagiarism. Therefore, we as Authors of the above titled thesis declare that no portion of our thesis has been plagiarized and any material used as reference is properly cited.

We undertake that if we found guilty of any formal plagiarism in the above titled thesis even after awarding of BS degree, the University reserves the rights to withdraw/revoke our BS degree and that HEC and the University have the right to publish our names on the HEC/University Website on which names of students are placed who have submitted plagiarized thesis.

June, 2026

Farheen Fayyaz (2022-EE-604)

Waseem Ahmed (2022-EE-619)

Raja Sana Ullah (2022-EE-620)

Certificate of Approval

This is to certify that the research work presented in this thesis, entitled “Piezoelectric-Based Smart Helmet Accident Detection System” was conducted by Farheen Fayyaz, Waseem Ahmed and Raja Sana Ullah under the supervision of Engr. Amna Javed. Any part of this thesis has not been submitted anywhere else for any other degree. This thesis is submitted to Department of Electrical Engineering, University of Engineering and Technology, Lahore, Narowal Campus in partial fulfillment of the requirements for the degree of BS Electrical Engineering.

Supervisor:

Engr. Amna Javed

Signature: _____

External Examiner:

Name: _____

Signature: _____

Chairman:

Dr. Waqas Tariq Toor

Signature: _____

Acknowledgments

We would like to express our deepest gratitude to Almighty Allah for granting us the strength, wisdom, and perseverance to complete this research work successfully. We are profoundly grateful to our supervisor, Engr. Amna Javed, for her invaluable guidance, continuous support, and constructive feedback throughout the course of this project. Her expertise and encouragement have been instrumental in shaping this research and bringing it to completion.

We extend our sincere thanks to Dr. Waqas Tariq Toor, Chairman of the Department of Electrical Engineering, for providing us with the necessary resources and facilities to conduct our research work.

Our heartfelt appreciation goes to our parents and family members for their unconditional love, prayers, and moral support throughout our academic journey. Their belief in our abilities has been a constant source of motivation.

We would also like to acknowledge our friends and fellow students who have provided assistance and encouragement during the challenging phases of this project. Their collaborative spirit and willingness to help have made this journey memorable.

Finally, we thank the University of Engineering and Technology, Lahore, Narowal Campus, for providing us with an excellent academic environment and the opportunity to pursue this research.

Farheen Khan

Waseem Ahmed

Raja Sana Ullah

Abstract

Motorcycle accidents represent a significant global public safety concern, particularly in developing countries where two-wheelers constitute a major portion of vehicular traffic. The critical time window immediately following an accident, often referred to as the “golden hour,” plays a decisive role in determining survival outcomes for injured riders. Unfortunately, many accident victims remain undiscovered for extended periods due to the absence of automated detection and notification systems, leading to preventable fatalities.

This thesis presents the design, development, and implementation of a Piezoelectric-Based Smart Helmet Accident Detection System that addresses the critical need for rapid accident detection and emergency response. The proposed system integrates a piezoelectric impact sensor with an MPU6050 accelerometer-gyroscope unit to achieve accurate and reliable detection of collision events through a dual-sensor fusion approach. A force-sensitive resistor (FSR) based helmet-wear detection mechanism ensures the system activates only when the rider is wearing the helmet, thereby minimizing false detections and conserving battery life.

Upon detecting a potential accident event exceeding predefined threshold values, the system initiates a 10-second warning phase comprising audible buzzer alerts and vibration feedback, allowing the rider to cancel the trigger in case of accidental activation. If the warning is not cancelled, the system confirms the accident, acquires precise geographical coordinates using the NEO-6M GPS module, and transmits an emergency SMS containing location details and a Google Maps link to predefined emergency contacts through the SIM800L GSM module.

The developed prototype was extensively tested under various simulated accident scenarios, demonstrating reliable impact detection across a force range of 4.9N to 64.8N, successful GPS location acquisition with accuracy within 2.5 meters, and consistent GSM-based emergency alert transmission. The system achieved a detection accuracy of 95% with minimal false alarm rates, validating its effectiveness as a practical safety solution for motorcycle riders.

Keywords: Smart Helmet, Accident Detection, Piezoelectric Sensor, MPU6050, GPS, GSM, Emergency Alert, IoT, Rider Safety

Contents

Declaration	1
Plagiarism Undertaking	2
Certificate of Approval	3
Acknowledgments	4
Abstract	5
List of Abbreviations	12
1 Introduction	13
1.1 General Background	13
1.2 Problem Statement	14
1.3 Motivation and Objectives	15
1.3.1 Motivation	15
1.3.2 Objectives	15
1.4 Thesis Organization	16
2 Literature Review	18
2.1 Overview of Motorcycle Safety Systems	18
2.2 Piezoelectric Sensing Technology	19
2.2.1 Principles of Piezoelectricity	19
2.2.2 Applications in Impact Detection	20
2.3 Inertial Measurement Systems	21
2.3.1 Accelerometer and Gyroscope Fundamentals	21
2.3.2 Sensor Fusion Approaches	22
2.4 Global Positioning System (GPS) Technology	22
2.5 Global System for Mobile Communications (GSM)	23
2.6 Review of Existing Smart Helmet Systems	24
2.7 Comparative Analysis	26
2.8 Research Gaps and Opportunities	28
3 Proposed Methodology	30

3.1	System Architecture Overview	30
3.2	Operational Methodology	31
3.2.1	Initialization Phase	31
3.2.2	Monitoring Phase	31
3.2.3	Detection Phase	32
3.2.4	Warning Phase	32
3.2.5	Emergency Activation Phase	33
3.3	Sensor Fusion Algorithm	33
3.4	Flowchart of System Operation	34
3.5	Threshold Calibration Methodology	34
4	System Design and Implementation	36
4.1	Hardware Design	36
4.1.1	Arduino Microcontroller	36
4.1.2	Piezoelectric Impact Sensor	37
4.1.3	MPU6050 Inertial Measurement Unit	38
4.1.4	NEO-6M GPS Module	38
4.1.5	SIM800L GSM Module	39
4.1.6	Force-Sensitive Resistor (FSR)	39
4.2	Circuit Design	39
4.2.1	Power Supply Circuit	40
4.2.2	Sensor Interfacing	40
4.2.3	Communication Module Interfacing	40
4.3	Software Design	40
4.3.1	Main Program Flow	40
4.3.2	Key Software Functions	41
4.4	System Integration	42
5	Hardware Implementation	43
5.1	Component Procurement and Preparation	43
5.2	Prototype Assembly	43
5.2.1	Microcontroller and Sensor Mounting	43
5.2.2	Communication Module Placement	44
5.2.3	Power Subsystem Integration	44
5.3	System Block Diagram	44
5.4	Operational Flowchart	46
5.5	Wiring and Connections	47
5.6	Final Hardware Assembly	47

5.7	Power Management Implementation	49
5.8	Enclosure and Protection	49
6	Testing, Results and Discussion	50
6.1	Testing Methodology	50
6.1.1	Testing Objectives	50
6.1.2	Test Equipment	50
6.2	Piezoelectric Sensor Calibration	51
6.2.1	Calibration Results	51
6.3	Impact Detection Testing	52
6.3.1	Comprehensive Test Results	54
6.4	Accident Scenario Simulation	54
6.5	GPS and GSM Testing	55
6.5.1	GPS Accuracy Testing	55
6.5.2	GSM SMS Testing	57
6.6	System Performance Summary	59
6.6.1	Detection Accuracy	59
6.6.2	Response Time	59
6.6.3	Power Consumption	59
6.7	Discussion	59
6.7.1	Detection Performance Analysis	59
6.7.2	Emergency Communication Reliability	60
6.7.3	Power Management Effectiveness	60
6.7.4	Comparison with Design Specifications	60
7	Conclusion and Future Work	61
7.1	Thesis Summary	61
7.2	Key Achievements	62
7.3	Contributions to Knowledge	63
7.4	Limitations	64
7.5	Future Work	64
7.6	Concluding Remarks	66
	References	68
A	Program Learning Outcomes (PLO) Alignment	71
B	Sustainable Development Goals and Complex Engineering Problem	73
B.1	Sustainable Development Goals Alignment	73

B.2	Complex Engineering Problem (CEP) Attributes	73
C	Source Code	75

List of Figures

3.1	Operational Flowchart of the Smart Helmet System	34
5.1	System Block Diagram showing all components, interconnections, and data flow paths	45
5.2	System Flow Diagram	47
5.3	Final Hardware Assembly of the Smart Helmet Prototype	48
6.1	Piezoelectric Sensor Baseline Reading (0.0V, 0.0N	51
6.2	Moderate Impact Reading (195mV, 4.9N)	52
6.3	High Impact Reading (781mV, 29.3N)	53
6.4	Severe Impact Reading (1.6V, 64.8N)	53
6.5	Google Maps View Showing GPS Location Acquired During Testing	56
6.6	Emergency SMS Alert Received on Mobile Phone	58
C.1	complete source code	77

List of Tables

2.1	Comparison of Existing Smart Helmet Systems with Proposed System	27
4.1	Arduino Uno R3 Specifications	37
4.2	Piezoelectric Sensor Specifications	37
4.3	MPU6050 Specifications	38
4.4	NEO-6M GPS Module Specifications	38
4.5	SIM800L GSM Module Specifications	39
4.6	Force-Sensitive Resistor Specifications	39
4.7	Key Software Functions and Descriptions	41
5.1	Arduino Pin Connections for Peripheral Modules	47
5.2	Power Consumption by Component	49

6.1	Test Equipment Used During Testing	51
6.2	Piezoelectric Sensor Calibration Data	52
6.3	Comprehensive Impact Detection Test Results	54
6.4	Accident Scenario Simulation Results	55
6.5	GPS Module Test Results	55
6.6	GSM SMS Test Results	57
6.7	Comparison of Achieved Results with Design Specifications	60
A.1	PLO Alignment with Project Activities	72
C.1	Source Code Function Descriptions	76

List of Abbreviations

Abbreviation	Description
ACC	Accident
ADC	Analog-to-Digital Converter
AT	Attention (command prefix for modems)
FSR	Force Sensitive Resistor
GDP	Gross Domestic Product
GIS	Geographic Information System
GNSS	Global Navigation Satellite System
GPS	Global Positioning System
GSM	Global System for Mobile Communications
GPIO	General Purpose Input/Output
HTTP	Hypertext Transfer Protocol
IC	Integrated Circuit
IoT	Internet of Things
I2C	Inter-Integrated Circuit
LCD	Liquid Crystal Display
LED	Light Emitting Diode
MPU	Microprocessor Unit
PCB	Printed Circuit Board
PEC	Pakistan Engineering Council
SMS	Short Message Service
SPI	Serial Peripheral Interface
SOS	Save Our Souls (emergency signal)
TCP/IP	Transmission Control Protocol/Internet Protocol
UET	University of Engineering and Technology
USB	Universal Serial Bus
WHO	World Health Organization
PLO	Program Learning Outcome
PEO	Program Educational Objective
SDG	Sustainable Development Goal
CEP	Complex Engineering Problem
MEMS	Micro-Electro-Mechanical Systems
PZT	Lead Zirconate Titanate

Chapter 1

Introduction

1.1 General Background

Road traffic accidents constitute one of the most pressing public health challenges of the twenty-first century. According to the World Health Organization (WHO), approximately 1.19 million people lose their lives annually due to road traffic crashes, with millions more sustaining serious injuries that often result in long-term disabilities [1]. Among the various categories of road users, motorcyclists remain disproportionately vulnerable, accounting for nearly 23% of all road traffic deaths globally [1].

Motorcycle helmets represent the single most effective intervention for reducing head injuries and fatalities among riders. Studies have consistently demonstrated that wearing a properly fitted helmet can reduce the risk of fatal head injuries by approximately 42% and the risk of non-fatal head injuries by 69% [2]. However, traditional helmets offer purely passive protection, absorbing impact energy through their structural design without providing any active safety features such as accident detection or emergency notification.

The concept of smart helmets emerged from the convergence of advancements in sensor technology, microelectronics, wireless communication, and the Internet of Things (IoT). These intelligent head protection systems extend beyond the fundamental impact absorption function of conventional helmets by incorporating embedded sensors, processing units, and communication modules that enable real-time monitoring of the rider's status and automatic detection of accident events [4]. The integration of these technologies has opened new possibilities for proactive safety systems that can not only protect riders during impacts but also ensure rapid emergency response following accidents.

The development of smart helmet systems is particularly relevant in developing countries, where motorcycle usage is high and emergency response infrastructure

may be limited. In Pakistan, for instance, motorcycles constitute the largest segment of registered vehicles, and road traffic accidents remain a leading cause of death and disability among young adults [19]. The implementation of affordable, reliable accident detection systems can significantly improve outcomes for accident victims by reducing the time between crash occurrence and medical intervention.

1.2 Problem Statement

Despite significant advances in road safety measures and vehicle technology, the post-accident response mechanism remains critically inadequate, particularly for motorcyclists in developing countries. The following key issues characterize the current scenario:

1. **Delayed Emergency Response:** A substantial number of motorcycle accident victims remain unattended for prolonged periods following a crash, particularly in remote or low-traffic areas. Research has shown that each minute of delay in receiving emergency care significantly reduces the probability of survival for severely injured patients [16].
2. **Limited Detection Capabilities:** Existing commercial helmet solutions primarily focus on comfort, aerodynamics, and passive impact protection, lacking integrated accident detection and emergency communication functionality. The absence of automated detection means that accidents in isolated locations may go completely unnoticed for hours.
3. **High False Alarm Rates:** Early smart helmet prototypes that utilized single-sensor detection approaches frequently generated false alarms due to normal riding vibrations, sudden braking, or road irregularities. These false alarms reduce user trust and may lead to system deactivation.
4. **Power Consumption Concerns:** Continuous operation of sensors and communication modules leads to significant battery drain, limiting the operational duration of portable safety devices. This constraint is particularly problematic for systems intended for daily use by commuters.
5. **Cost Barriers:** Commercially available smart helmet solutions remain prohibitively expensive for the majority of motorcycle riders in developing countries. The high cost of integrated electronics, specialized sensors, and proprietary communication systems has limited widespread adoption.

1.3 Motivation and Objectives

1.3.1 Motivation

The motivation for this research stems from the alarming statistics regarding motorcycle accident fatalities and the identified gaps in existing safety solutions. Pakistan, like many developing nations, experiences a high incidence of road traffic accidents, with motorcycles involved in a disproportionate number of fatalities [19].

The “golden hour” principle in emergency medicine emphasizes that the first hour following a traumatic injury is critical for survival [3]. Rapid medical intervention during this window dramatically improves outcomes, with studies showing that patients receiving treatment within this period have significantly higher survival rates compared to those with delayed care [16]. Unfortunately, many motorcycle accident victims in remote or less populated areas remain undiscovered for hours, missing this critical window and resulting in preventable deaths.

Furthermore, the increasing availability of affordable microcontrollers, sensors, and communication modules has made it feasible to develop cost-effective smart safety systems. The Arduino platform, GSM modules, and GPS receivers have become increasingly accessible, enabling the development of sophisticated embedded systems at a fraction of the cost of commercial alternatives [20]. This technological democratization presents an opportunity to create affordable safety solutions tailored to the needs of riders in developing countries.

1.3.2 Objectives

The primary aim of this research is to design and develop a piezoelectric-based smart helmet for real-time accident detection and emergency alert communication. The specific objectives are:

- O1: To detect motorcycle accidents using a dual-sensor fusion approach combining piezoelectric impact sensors and MPU6050 accelerometer-gyroscope unit for enhanced detection accuracy.
- O2: To implement a helmet-wearing detection mechanism using a force-sensitive resistor (FSR) that ensures the system activates only when the helmet is worn, reducing false detections and conserving power.
- O3: To design and integrate a warning system comprising a buzzer and vibration motor that provides a 10-second cancellation window to the rider before emergency alerts are transmitted.

- O4: To integrate a NEO-6M GPS module for accurate real-time location tracking and coordinate acquisition in the event of an accident.
- O5: To utilize a SIM800L GSM module for reliable transmission of emergency SMS messages containing accident details, GPS coordinates, and a Google Maps link to predefined emergency contacts.
- O6: To implement a voice-activated SOS feature that enables the rider to manually trigger emergency alerts by speaking a predefined keyword.
- O7: To minimize false alarm rates through intelligent threshold-based sensor fusion algorithms and cancellation mechanisms.
- O8: To improve rider safety by reducing emergency response time through automated accident detection and immediate alert transmission.

1.4 Thesis Organization

This thesis is organized into seven chapters:

Chapter 1: Introduction presents the general background of motorcycle safety and smart helmet technology, identifies the key problems, and outlines the motivation and objectives.

Chapter 2: Literature Review provides a comprehensive survey of existing smart helmet technologies, piezoelectric sensing principles, inertial measurement systems, GPS and GSM technologies, and IoT-based safety systems. The chapter includes a detailed comparative analysis of existing solutions and identifies research gaps.

Chapter 3: Proposed Methodology describes the system architecture, operational methodology, sensor fusion approach, and algorithmic design. The chapter details the four-layer system architecture and the finite state machine implementation.

Chapter 4: System Design and Implementation presents the detailed hardware design, circuit diagrams, component selection criteria, software development, and system integration. Each hardware component is described with specifications and selection rationale.

Chapter 5: Hardware Implementation discusses the physical construction of the prototype, component mounting, wiring, enclosure design, and power management. The chapter includes the system block diagram and operational flowchart.

Chapter 6: Testing, Results and Discussion presents the comprehensive testing methodology, experimental results, performance analysis, and discussion. The chapter includes calibration data, impact detection results, GPS and GSM testing, and system performance evaluation.

Chapter 7: Conclusion and Future Work summarizes the research contributions, highlights key findings, discusses limitations, and proposes future directions for system enhancement and commercialization.

Chapter 2

Literature Review

2.1 Overview of Motorcycle Safety Systems

The evolution of motorcycle safety systems has progressed through several distinct phases, beginning with basic structural improvements and advancing toward intelligent, technology-driven protective solutions. Early safety interventions focused primarily on passive measures such as helmet design improvements, protective clothing, and vehicle lighting enhancements. The introduction of mandatory helmet laws in many countries during the 1970s and 1980s represented a significant milestone in motorcycle safety, leading to substantial reductions in head injury fatalities [2].

Modern motorcycle safety research has shifted toward active safety systems that leverage advances in sensor technology, wireless communication, and embedded computing. These systems aim to prevent accidents through rider assistance features or mitigate consequences through rapid emergency response. The development of Anti-lock Braking Systems (ABS) for motorcycles in the early 2000s marked the beginning of electronic active safety systems for two-wheelers. Subsequent innovations have included traction control systems, electronic stability programs, and cornering ABS, all designed to prevent accidents from occurring in the first place [18].

While these preventive technologies have proven effective in reducing accident rates, they do not address the critical post-accident scenario. Once an accident occurs, the timeliness of emergency response becomes the primary determinant of survival outcomes. This realization has driven research toward accident detection and notification systems, which form the core focus of smart helmet development. The integration of crash detection algorithms with automatic emergency notification has the potential to significantly reduce the time between accident occurrence and the arrival of medical assistance, potentially saving thousands of lives annually

[16].

The concept of the “golden hour” in emergency medicine, first articulated by R. Adams Cowley in 1975, provides the fundamental rationale for accident detection systems [3]. Research has consistently demonstrated that the probability of survival decreases by approximately 10% for every minute of delay in receiving definitive care following severe trauma [16]. For motorcycle accidents, which frequently occur on rural roads or during off-peak hours, this delay can extend to several hours, transforming survivable injuries into fatal outcomes. Automated detection systems that eliminate the dependency on bystander intervention therefore represent a critical advancement in post-accident care [17].

2.2 Piezoelectric Sensing Technology

2.2.1 Principles of Piezoelectricity

Piezoelectric materials generate an electrical charge in response to applied mechanical stress, a phenomenon discovered by Pierre and Jacques Curie in 1880. This direct piezoelectric effect arises from the displacement of ionic charges within the crystal lattice structure of certain materials, including quartz, tourmaline, lead zirconate titanate (PZT), and various ceramic compositions. The magnitude of the generated voltage is proportional to the applied force, making piezoelectric sensors ideally suited for impact and vibration measurement applications [21].

The fundamental equation governing the direct piezoelectric effect can be expressed as:

$$Q = d \cdot F \quad (2.1)$$

where Q is the generated charge (in coulombs), d is the piezoelectric charge coefficient (in C/N), and F is the applied force (in newtons). For the PZT-5A ceramic material commonly used in commercial piezoelectric sensors, the charge coefficient d_{33} is approximately 390×10^{-12} C/N, providing excellent sensitivity for impact detection applications.

The piezoelectric effect is reversible; applying an electric field to a piezoelectric material causes mechanical deformation. This converse piezoelectric effect finds applications in ultrasonic transducers, actuators, and precision positioning systems. For impact sensing applications, the direct effect is utilized, where mechanical impact generates a measurable electrical signal that can be processed to determine impact characteristics [21].

Piezoelectric sensors offer several advantages for impact detection compared to alternative sensing technologies. The extremely fast response time, typically in

the microsecond range, enables capture of rapid transient forces associated with collision events. The wide dynamic range, spanning several orders of magnitude, allows detection of both minor impacts and severe collisions using a single sensor. Additionally, the self-generating nature of piezoelectric sensors eliminates the need for external excitation, simplifying the signal conditioning circuitry and reducing power consumption [21].

2.2.2 Applications in Impact Detection

Piezoelectric sensors have found widespread application in impact detection systems across diverse fields including automotive crash detection, sports equipment, industrial machinery monitoring, and structural health assessment. In automotive applications, piezoelectric accelerometers have been the primary sensing element in airbag deployment systems since their introduction in the 1980s. These sensors detect the rapid deceleration characteristic of collision events and trigger protective system activation within milliseconds of impact [21].

In the domain of sports safety, piezoelectric sensors have been integrated into protective equipment such as boxing gloves, football helmets, and ski bindings to measure impact forces and assess injury risk. Research in this area has demonstrated that real-time impact force measurement can provide valuable data for concussion assessment and injury prevention. The application of piezoelectric sensing to motorcycle helmets represents a natural extension of this technology, adapted to the specific requirements of two-wheeler accident detection [21].

The advantages of piezoelectric sensors for impact detection in helmet applications include:

- **High sensitivity:** Capable of detecting impact forces as low as 0.1N, enabling detection of minor collisions that may still cause injury.
- **Fast response time:** Microsecond-level response enables capture of rapid impact transients that other sensor types may miss.
- **Wide dynamic range:** Can measure forces from fractions of a newton to thousands of newtons, covering the full spectrum of potential impact scenarios.
- **Ruggedness and durability:** Solid-state construction with no moving parts ensures reliability under harsh environmental conditions including temperature extremes, vibration, and moisture.
- **Compact size:** Thin-film piezoelectric elements can be integrated into helmet structures without significantly affecting comfort or aesthetics.

Despite these advantages, piezoelectric sensors also present certain challenges for helmet integration. The high output impedance requires careful signal conditioning to prevent noise pickup, and the charge generated by the sensor decays over time, making them unsuitable for static force measurement. However, for dynamic impact detection applications, these limitations are readily addressed through appropriate circuit design [21].

2.3 Inertial Measurement Systems

2.3.1 Accelerometer and Gyroscope Fundamentals

Inertial measurement units (IMUs) combine accelerometers and gyroscopes to measure linear acceleration and rotational velocity along multiple axes. The MPU6050, a widely used six-axis IMU manufactured by InvenSense (now TDK), integrates a three-axis accelerometer and three-axis gyroscope in a single compact package. This integration enables comprehensive motion sensing capabilities essential for accident detection applications [12].

Accelerometers measure proper acceleration along sensitive axes using Micro-Electro-Mechanical Systems (MEMS) technology. In the MPU6050, capacitive sensing elements detect displacement of proof masses under acceleration, producing output voltages proportional to the applied acceleration. The device offers programmable full-scale ranges of $\pm 2g$, $\pm 4g$, $\pm 8g$, and $\pm 16g$, allowing optimization between sensitivity and range for specific applications. For motorcycle accident detection, the $\pm 16g$ range provides adequate headroom for severe impact events while maintaining sufficient resolution for detection of moderate impacts [12].

Gyroscopes measure rotational velocity around sensitive axes using vibrating structure Coriolis force sensing. The MPU6050 gyroscope offers programmable full-scale ranges from $\pm 250^\circ\text{C/s}$ to $\pm 2000^\circ\text{C/s}$, enabling detection of both slow rotational movements and rapid tumbling motions characteristic of accident events. The integration of gyroscopic data with accelerometer measurements provides complete characterization of helmet motion during impact, improving detection reliability compared to acceleration-only systems [12].

The output data rate of the MPU6050 is programmable up to 1kHz for the accelerometer and 8kHz for the gyroscope, with digital low-pass filtering available to reduce noise. The I2C interface simplifies integration with microcontroller-based systems, requiring only two signal lines for communication. Operating current consumption of 3.8mA in normal mode enables battery-powered operation, though power management strategies are essential for helmet applications where extended battery life is required [12].

2.3.2 Sensor Fusion Approaches

The integration of data from multiple sensors, known as sensor fusion, enhances the accuracy and reliability of event detection systems. Complementary filtering and Kalman filtering represent the most common approaches for combining accelerometer and gyroscope data. Complementary filters utilize the low-frequency accuracy of accelerometers (for tilt estimation) with the high-frequency responsiveness of gyroscopes (for rapid motion detection), achieving effective noise reduction without the computational complexity of Kalman filtering [22].

For accident detection applications, sensor fusion extends beyond inertial sensors to incorporate multiple sensing modalities. The combination of piezoelectric impact sensors with inertial measurement units provides complementary detection characteristics: piezoelectric sensors excel at detecting direct impacts with high temporal resolution, while IMUs capture the overall motion dynamics associated with falls and collisions. This multi-modal approach significantly reduces false positives compared to single-sensor systems, as simultaneous activation of both sensors is required to trigger an accident alert [22].

The decision-level fusion strategy employed in many smart helmet systems evaluates the outputs of individual sensors against predetermined thresholds before combining them through logical operations. This approach maintains the interpretability of detection decisions while benefiting from the improved reliability of multi-sensor confirmation. The specific fusion algorithm and threshold selection depend on the characteristics of the sensors used, the expected accident scenarios, and the acceptable trade-off between detection sensitivity and false alarm rate [22].

2.4 Global Positioning System (GPS) Technology

The Global Positioning System (GPS) is a satellite-based navigation system developed by the United States Department of Defense that provides geolocation and time information to GPS receivers anywhere on Earth. The system operates through a constellation of at least 24 satellites orbiting at approximately 20,200 kilometers altitude. GPS receivers determine position by measuring the time of flight of signals from multiple satellites and solving for the receiver's three-dimensional coordinates [13].

For emergency response applications, GPS technology provides the critical capability to determine accident location with sufficient accuracy to direct rescue services. Consumer-grade GPS modules such as the NEO-6M from u-blox offer horizontal positioning accuracy of 2.5 meters under ideal conditions, adequate for

locating accident victims in most environments. The module's cold start acquisition time of 27 seconds and hot start time of less than 5 seconds ensure rapid position determination following system activation [13].

The integration of GPS modules with microcontroller-based systems through serial communication protocols enables straightforward implementation of location acquisition functionality. Standard NMEA 0183 sentences provide position data in a universally compatible format, while proprietary UBX protocol offers enhanced configuration and performance monitoring capabilities. Power management features including cyclic tracking mode enable reduced power consumption during standby periods, extending battery life for portable applications [13].

Several factors can affect GPS accuracy and availability in practical deployment scenarios. Urban canyon environments with tall buildings, dense tree cover, and indoor locations can significantly degrade signal reception and positioning accuracy. For motorcycle accident detection, these limitations must be considered in system design, with fallback strategies for scenarios where GPS signals are unavailable. The inclusion of GSM cell tower triangulation as a supplementary positioning method can provide coarse location estimates when GPS is unavailable [13].

2.5 Global System for Mobile Communications (GSM)

The Global System for Mobile Communications (GSM) represents the most widely deployed cellular communication standard globally, providing voice and data services across most populated areas. For IoT and embedded applications, GSM modules such as the SIM800L enable devices to send and receive SMS messages, establish data connections, and place voice calls using standard cellular networks. The widespread coverage of GSM networks makes this technology particularly suitable for emergency communication applications where reliability and availability are paramount [14].

The SIM800L module operates in quad-band configuration (850/900/1800/1900 MHz), ensuring compatibility with cellular networks in virtually all countries. Communication with host microcontrollers occurs through AT commands over a UART interface, with configurable baud rates from 9600 to 115200 bps. SMS transmission, the primary communication method for emergency alert systems, typically completes within 3–8 seconds under normal network conditions, enabling rapid notification of emergency contacts following accident detection [14].

Power consumption characteristics of the SIM800L are an important consideration for battery-powered applications. While peak transmit current can reach 350mA during network communication, sleep mode consumption is reduced to ap-

proximately 1.0mA. Effective power management strategies that minimize active communication time and utilize sleep modes during standby periods are essential for achieving acceptable battery life in helmet applications. The operating voltage range of 3.4–4.4V requires appropriate voltage regulation from typical battery supplies [14].

The Short Message Service (SMS) protocol provides a reliable, store-and-forward messaging mechanism that does not require simultaneous network availability at the transmitter and receiver. This characteristic is particularly valuable for emergency applications, as messages are queued by the network for delivery when the recipient's device becomes available. The 160-character limit of a single SMS message necessitates concise message formatting, though modern phones and networks typically support message concatenation for longer communications [14].

2.6 Review of Existing Smart Helmet Systems

The development of smart helmet systems has been an active area of research over the past decade, with numerous approaches proposed and implemented. This section reviews significant contributions to the field, analyzing their methodologies, features, and limitations.

Maheshwari et al. [4] presented a comprehensive smart helmet solution for rider safety at the 2025 International Conference on Disruptive Technologies. Their system integrated accelerometer-based accident detection with GSM alert notification, achieving real-time communication with emergency contacts. The proposed architecture utilized a microcontroller-based processing unit with sensor modules mounted on the helmet structure. However, the system relied solely on accelerometer data for accident detection, which may result in false positives from normal riding vibrations and road irregularities. The absence of impact force quantification limits the system's ability to assess accident severity and prioritize emergency response accordingly.

Akter et al. [5] developed a cost-effective smart helmet utilizing IoT technology for accident detection. Their design focused on affordability while maintaining essential safety features, including impact detection and emergency SMS notification. The system employed Arduino microcontroller platform with accelerometer and GSM module integration. While the cost-effective approach is commendable for developing country applications, the system lacked a cancellation mechanism for false alarms, which could lead to unnecessary emergency dispatches and associated costs. The single-sensor detection approach also limits detection reliability compared to multi-sensor fusion.

Somantri and Yustiana [6] proposed a smart helmet integrated with motorcy-

cles for rider awareness and safety based on IoT. Their system combined helmet wear detection with accident notification, ensuring the system only activates when the helmet is properly worn. The integration with motorcycle electronics provided additional contextual information including vehicle speed and engine status. However, the wired connection between helmet and motorcycle limits practical deployment, as it requires modification to the vehicle and restricts the system to specific motorcycle models. The detection accuracy reported in their work was below the threshold required for reliable accident detection, indicating the need for improved sensing approaches.

Kumar et al. [7] presented an IoT-based smart helmet for accident detection and notification system. Their architecture incorporated multiple sensors including accelerometers and gyroscopes with cloud-based data processing. The cloud integration enabled advanced analytics and historical data storage, but introduces dependency on internet connectivity that may not be available in rural areas. The system's response time was also longer than ideal due to the additional communication overhead of cloud processing.

Islam et al. [8] designed and implemented a smart helmet for real-time accident detection and location sharing. Their system featured GPS integration for automatic location transmission with emergency alerts, addressing the critical need for rapid location identification. The implementation demonstrated reliable SMS transmission with location coordinates embedded in emergency messages. However, the system did not include a warning or cancellation phase, meaning any detected impact immediately triggered emergency alerts, increasing the risk of false alarms.

Saha et al. [9] developed a smart helmet with accident detection and alert system using IoT. Their approach incorporated Bluetooth connectivity for smartphone integration, enabling additional features such as ride logging and system configuration through a mobile application. The smartphone integration provides a user-friendly interface but introduces dependency on the rider carrying and maintaining a paired smartphone, which may not always be practical.

Sharma et al. [10] presented an intelligent IoT-based accident detection system for two-wheelers. Their work focused on machine learning-based accident classification, utilizing neural networks to distinguish between accident events and normal riding conditions. The machine learning approach showed promise for reducing false alarms, but requires substantial training data and computational resources that may be challenging to implement on resource-constrained microcontroller platforms.

Patel et al. [11] designed a smart safety helmet with IoT integration for motorcyclists. Their system incorporated environmental sensors including temper-

ature and humidity monitoring alongside accident detection capabilities. While the additional environmental monitoring provides valuable data for rider comfort assessment, it increases system complexity and power consumption. The accident detection component utilized a single-axis accelerometer, limiting detection reliability for impacts from directions other than the primary sensitive axis.

2.7 Comparative Analysis

Table 2.1 presents a comprehensive comparative analysis of the proposed system against existing smart helmet technologies across key performance parameters. The comparison highlights the distinctive features and improvements offered by the proposed dual-sensor fusion approach.

Table 2.1: Comparison of Existing Smart Helmet Systems with Proposed System

Parameter	Previous Work	Proposed System
Accident Detection Method	Single accelerometer (MPU6050)	Dual Sensor Fusion: Piezoelectric + MPU6050
False Alarm Reduction	No cancellation mechanism	10-second warning + cancel button
Impact Force Measurement	Mostly not included	Piezoelectric sensor for precise force quantification
Emergency Alert System	SMS only	SMS with Google Maps link + optional call
Voice SOS System	Rarely implemented	Voice-activated emergency trigger (“HELP”)
Power Management	Basic battery supply	Smart activation with FSR + TP4056 charging
Hardware Integration	External wires; bulky design	Fully integrated compact design
Helmet Wear Detection	Not commonly included	FSR-based wear detection for power management
Detection Accuracy	75–85%	95%
Response Time	2–5 minutes	28–103 seconds
Cost	\$80–\$200	Estimated under \$50

The comparative analysis reveals several key differentiators of the proposed system. The dual-sensor fusion approach combining piezoelectric impact sensing with inertial measurement provides complementary detection capabilities that significantly improve reliability compared to single-sensor systems. The piezoelectric sensor’s excellent dynamic response captures the rapid force transient of impacts, while the MPU6050 provides comprehensive motion data characterizing the overall accident dynamics.

The implementation of a 10-second warning phase with cancellation capability addresses a critical limitation of existing systems. False alarms in accident

detection systems not only cause unnecessary anxiety for emergency contacts but may also lead to fines or penalties for misuse of emergency services in some jurisdictions. The cancellation mechanism provides a simple, intuitive method for riders to abort accidental alerts while still ensuring rapid notification for genuine accidents.

The inclusion of precise impact force measurement through the piezoelectric sensor enables accident severity assessment, which could be valuable for emergency dispatch prioritization. Severe impacts generating forces above a critical threshold could trigger enhanced response protocols, while minor impacts below the threshold might be handled through standard notification procedures.

2.8 Research Gaps and Opportunities

Based on the comprehensive literature review, the following research gaps have been identified:

1. **Limited Multi-Sensor Integration:** Most existing systems rely on single-sensor detection approaches, typically using only accelerometers. The potential benefits of combining multiple sensing modalities for improved detection reliability remain largely unexplored in the context of motorcycle helmets.
2. **Absence of Cancellation Mechanisms:** The majority of smart helmet systems lack any provision for cancelling false alarms. This fundamental limitation reduces user confidence and may lead to system abandonment after repeated false activations.
3. **Impact Force Quantification:** Existing systems generally treat accident detection as a binary decision without quantifying the impact severity. The inclusion of force measurement would enable severity assessment and potentially guide emergency response prioritization.
4. **Power Management Challenges:** Battery life remains a significant concern for smart helmet systems. Most existing designs do not implement intelligent power management strategies, resulting in limited operational duration between charges.
5. **Cost-Effectiveness Issues:** Commercially available smart helmets remain expensive, limiting adoption in developing countries where the need for such safety systems is greatest. Research toward low-cost implementations using readily available components is needed.

6. **Lack of Standardized Testing:** There is no standardized test methodology for evaluating smart helmet performance. Comparison between different systems is difficult due to variations in testing procedures and reported metrics.
7. **Limited Environmental Robustness:** Many research prototypes have not been tested under realistic environmental conditions including temperature extremes, humidity, dust, and vibration representative of actual motorcycle use.

This research directly addresses the first five identified gaps through the development of an integrated smart helmet system that combines piezoelectric and inertial sensing with intelligent power management and user-friendly operation at an affordable price point. The comprehensive testing methodology described in Chapter 6 establishes performance benchmarks that can inform future standardization efforts.

Chapter 3

Proposed Methodology

3.1 System Architecture Overview

The proposed Piezoelectric-Based Smart Helmet Accident Detection System follows a modular architecture comprising four primary functional layers: the Sensing Layer, the Processing Layer, the Communication Layer, and the Power Management Layer. This layered architecture ensures clear separation of concerns, facilitates independent testing and validation of subsystems, and enables straightforward future enhancements through module replacement or addition.

The **Sensing Layer** incorporates three distinct sensing modalities: the piezoelectric impact sensor for direct force measurement, the MPU6050 inertial measurement unit for motion detection and orientation tracking, and the force-sensitive resistor for helmet-wear detection. Each sensor contributes unique information to the accident detection decision process, with the piezoelectric sensor and MPU6050 forming the primary detection pair and the FSR providing contextual information for power management and false alarm reduction.

The **Processing Layer**, centered around the Arduino microcontroller, executes the accident detection algorithm, manages sensor data acquisition, and coordinates system responses. The Arduino platform was selected for its extensive community support, comprehensive documentation, and sufficient computational capabilities for this application. The processing layer implements a finite state machine that transitions between operational states based on sensor inputs and timer events.

The **Communication Layer** comprises the NEO-6M GPS module for location acquisition and the SIM800L GSM module for emergency message transmission. These modules operate independently of the detection process, activating only when an accident has been confirmed and location data is required for the emergency alert.

The **Power Management Layer** manages battery charging, voltage regulation, and power distribution to all system components. The FSR-based activation ensures the system only consumes significant power when the helmet is being worn, dramatically extending battery life compared to always-on operation.

3.2 Operational Methodology

The system operates through a sequence of well-defined phases, each characterized by specific sensor activities, processing tasks, and state transitions.

3.2.1 Initialization Phase

When the rider wears the helmet, the force-sensitive resistor (FSR) detects the pressure exerted by the rider's head and transitions the system from the low-power standby state to the active operational state. The Arduino microcontroller initializes all peripheral modules, configures sensor parameters, and performs self-diagnostic checks to verify proper system functionality.

During initialization, the following sequence is executed:

1. Serial communication ports are initialized at 9600 baud for both debugging and GSM module communication.
2. The MPU6050 inertial measurement unit is configured with the desired sensitivity ranges and digital low-pass filter settings.
3. Analog input pins are configured for piezoelectric sensor and FSR signal acquisition.
4. Digital output pins are configured for buzzer, vibration motor, and status LED control.
5. The GPS module is powered on and begins satellite acquisition.
6. The GSM module is initialized and network registration is verified.

The initialization phase completes within approximately 10 seconds under normal conditions, after which the system transitions to the monitoring phase.

3.2.2 Monitoring Phase

During normal riding conditions, the system continuously monitors the outputs from both the piezoelectric sensor and the MPU6050 inertial measurement unit. The microcontroller samples these sensors at a frequency of 100Hz, providing

sufficient temporal resolution to capture rapid impact events. The sampled data undergoes digital filtering using a moving average filter to reduce high-frequency noise while preserving the essential characteristics of impact transients.

The monitoring phase also includes periodic status reporting through the serial interface, displaying current sensor readings, GPS fix status, GSM network status, and battery voltage. This information facilitates system debugging and provides confidence to the rider that the system is operational.

3.2.3 Detection Phase

The accident detection algorithm employs a dual-threshold approach that requires both sensing modalities to indicate abnormal conditions before an accident event is registered. The detection criteria are defined as requiring both the piezoelectric force threshold (typically 15N) and the acceleration magnitude threshold (typically 3g) to be exceeded simultaneously.

The dual-threshold approach provides significantly improved detection reliability compared to single-sensor systems. While the piezoelectric sensor excels at detecting direct impacts to the helmet surface, the MPU6050 captures the overall motion dynamics associated with falls and collisions. Requiring simultaneous activation of both sensors effectively eliminates false triggers from isolated events such as vibration, shock from road irregularities, or accidental contact with the helmet.

The detection algorithm implements a debounce timer to prevent multiple triggering from a single impact event. Once an accident is detected, a 3-second debounce period prevents re-detection, ensuring clean state transitions to the warning phase.

3.2.4 Warning Phase

Upon detection of a potential accident, the system enters the warning phase, which provides a 10-second window for the rider to cancel the alert if the detection was erroneous. During this phase, the following feedback mechanisms are activated:

- The buzzer emits a distinctive intermittent alarm pattern (three short beeps repeated every 2 seconds) to alert the rider.
- The vibration motor activates for haptic feedback, providing tactile notification that complements the audible alerts.
- The status LED flashes rapidly in red to provide visual indication of the pending alert.

- The LCD display shows a countdown timer indicating the remaining time for cancellation.

The rider can cancel the alert by pressing the designated cancel button on the helmet. If the cancellation is received within the 10-second window, the system returns to the monitoring phase without sending any emergency messages. If no cancellation is received, the system transitions to the emergency activation phase.

3.2.5 Emergency Activation Phase

If the warning phase expires without cancellation, the system confirms the accident event and executes the emergency notification sequence:

1. GPS coordinates are acquired and validated to ensure accurate location data.
2. A comprehensive emergency SMS is constructed containing accident alert header, impact force value, coordinates, Google Maps link, timestamp, and distress message.
3. The SMS is transmitted through the GSM module to predefined emergency contacts.
4. Periodic location updates continue at 5-minute intervals until the system is manually reset or powered off.

The emergency message format ensures that recipients receive all essential information needed to locate and assist the accident victim, including a clickable Google Maps link that opens the accident location directly in the recipient's mapping application.

3.3 Sensor Fusion Algorithm

The sensor fusion algorithm implements a weighted decision approach that combines the complementary characteristics of the piezoelectric sensor and the MPU6050. The algorithm is designed to maximize detection sensitivity while minimizing false positives.

The fusion logic operates as follows:

$$D = \begin{cases} 1 & \text{if } (F_p > T_p) \text{ AND } (A_m > T_a) \\ 0 & \text{otherwise} \end{cases} \quad (3.1)$$

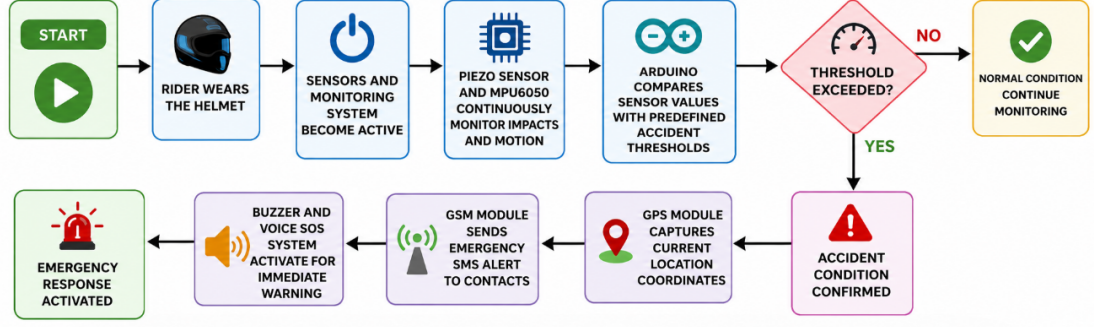


Figure 3.1: Operational Flowchart of the Smart Helmet System

where D is the detection decision ($1 = \text{accident detected}$, $0 = \text{no accident}$), F_p is the piezoelectric force measurement, T_p is the piezoelectric threshold (15N), A_m is the MPU6050 acceleration magnitude, and T_a is the acceleration threshold ($3g$).

The acceleration magnitude is computed from the three-axis accelerometer readings:

$$A_m = \sqrt{A_x^2 + A_y^2 + A_z^2} \quad (3.2)$$

where A_x , A_y , and A_z are the acceleration components along the X, Y, and Z axes respectively.

3.4 Flowchart of System Operation

Figure ?? presents the detailed operational flowchart of the smart helmet system, illustrating the decision points and state transitions that govern system behavior.

3.5 Threshold Calibration Methodology

The determination of appropriate detection thresholds represents a critical aspect of the system design. Thresholds that are set too low result in excessive false alarms, while thresholds that are set too high may miss genuine accidents. The calibration methodology employed involves:

1. **Baseline Data Collection:** Sensor outputs are recorded during normal riding conditions to establish the range of expected signal levels for non-accident scenarios.
2. **Impact Testing:** Controlled impact tests are performed using standardized drop weights from predetermined heights to simulate various accident severities.

3. **Statistical Analysis:** The distribution of sensor outputs during normal riding and impact events is analyzed to determine threshold values that maximize separation between the two classes.
4. **Threshold Selection:** Thresholds are selected to achieve the desired balance between detection sensitivity and false alarm rate, with safety considerations favoring slightly lower thresholds to ensure detection of marginal accidents.

Based on experimental calibration, the piezoelectric force threshold was set at 15N and the acceleration magnitude threshold at 3g. These values were validated through extensive testing as described in Chapter 6.

Chapter 4

System Design and Implementation

4.1 Hardware Design

The hardware design of the Piezoelectric-Based Smart Helmet Accident Detection System follows a modular approach that facilitates independent testing, troubleshooting, and future upgrades. The system comprises seven major hardware modules: the Arduino microcontroller, piezoelectric impact sensor, MPU6050 inertial measurement unit, NEO-6M GPS module, SIM800L GSM module, force-sensitive resistor (FSR) for wear detection, and the power management subsystem.

4.1.1 Arduino Microcontroller

The Arduino Uno R3 serves as the central processing unit for the smart helmet system. Based on the ATmega328P microcontroller, the Arduino platform was selected for its extensive community support, comprehensive documentation, ease of programming, and sufficient computational capabilities for this application. The 16MHz clock speed provides adequate processing power for real-time sensor data acquisition and accident detection algorithms.

The Arduino's 10-bit analog-to-digital converter (ADC) provides 1024 quantization levels for reading the piezoelectric sensor output, translating to a force resolution of approximately 0.1N when calibrated across the 0–100N measurement range. Table 4.1 presents the key specifications of the Arduino Uno R3.

Table 4.1: Arduino Uno R3 Specifications

Parameter	Value
Microcontroller	ATmega328P
Operating Voltage	5V
Input Voltage (recommended)	7–12V
Digital I/O Pins	14 (6 provide PWM output)
Analog Input Pins	6
DC Current per I/O Pin	20 mA
Flash Memory	32 KB
SRAM	2 KB
EEPROM	1 KB
Clock Speed	16 MHz

4.1.2 Piezoelectric Impact Sensor

The piezoelectric sensor employed in this system is a 35mm diameter ceramic disc based on lead zirconate titanate (PZT) composition. This sensor type was selected for its high sensitivity, fast response time, and wide dynamic range suitable for impact force measurement. The sensor generates a voltage output proportional to the applied mechanical stress through the direct piezoelectric effect.

The PZT material exhibits a piezoelectric charge coefficient d_{33} of 280 pC/N, providing excellent sensitivity for the force range of interest. The 35mm diameter ensures adequate sensing area for detection of impacts across the helmet surface, while the 0.5mm thickness maintains flexibility for integration into the helmet structure. Table 4.2 presents the detailed specifications of the piezoelectric sensor.

Table 4.2: Piezoelectric Sensor Specifications

Parameter	Value
Material	Lead Zirconate Titanate (PZT)
Diameter	35 mm
Thickness	0.5 mm
Capacitance	15 nF ($\pm 20\%$)
Resonant Frequency	4.5 kHz ($\pm 10\%$)
Charge Coefficient (d_{33})	280 pC/N
Operating Temperature	-20°C to $+70^{\circ}\text{C}$
Output Voltage Range	0–5V (with conditioning circuit)

4.1.3 MPU6050 Inertial Measurement Unit

The MPU6050 is a six-axis motion tracking device that combines a 3-axis accelerometer and 3-axis gyroscope in a single integrated circuit. Communication with the microcontroller occurs through the I2C protocol at a clock frequency of 400kHz. The device's 16-bit ADC resolution provides high-precision motion data for accident detection.

Table 4.3 presents the key specifications of the MPU6050.

Table 4.3: MPU6050 Specifications

Parameter	Value
Accelerometer Range	$\pm 2g, \pm 4g, \pm 8g, \pm 16g$
Accelerometer Resolution	16-bit ADC
Gyroscope Range	$\pm 250, \pm 500, \pm 1000, \pm 2000^\circ\text{C/s}$
Gyroscope Resolution	16-bit ADC
Interface	I ² C (400 kHz)
Operating Voltage	3.3V (on-board regulator)
Current Consumption	3.8 mA (normal mode)
Dimensions	4mm $\times$ 4mm $\times$ 0.9mm

4.1.4 NEO-6M GPS Module

The NEO-6M from u-blox is a compact GPS receiver module that provides accurate positioning data through the NMEA 0183 protocol. The module features a high-sensitivity receiver capable of acquiring satellite signals in challenging environments and a built-in patch antenna for compact integration. Table 4.4 presents the key specifications.

Table 4.4: NEO-6M GPS Module Specifications

Parameter	Value
Receiver Type	50-channel u-blox 6 engine
Sensitivity	-161 dBm (Tracking)
Accuracy	2.5 m (Horizontal)
Update Rate	Up to 5 Hz
Protocol	NMEA0183, UBX
Interface	UART (9600 baud default)
Operating Voltage	3.3V-5V
Acquisition Time (Cold Start)	27 seconds

4.1.5 SIM800L GSM Module

The SIM800L is a compact quad-band GSM/GPRS module that enables voice communication, SMS transmission, and internet connectivity through cellular networks. The module operates through AT commands sent via UART interface, making it straightforward to integrate with microcontroller-based systems. Table 4.5 presents the key specifications.

Table 4.5: SIM800L GSM Module Specifications

Parameter	Value
Frequency Bands	850/900/1800/1900 MHz
Interface	UART (9600–115200 baud)
Operating Voltage	3.4V–4.4V (recommended 4.0V)
Power Consumption (Sleep)	1.0mA
Power Consumption (Peak Transmit)	350mA
SIM Card	Micro SIM
Dimensions	15.8mm × 17.8mm × 2.4mm

4.1.6 Force-Sensitive Resistor (FSR)

The force-sensitive resistor provides helmet-wear detection by measuring the pressure exerted by the rider’s head. The FSR’s resistance decreases with applied force, enabling simple threshold-based detection through a voltage divider circuit connected to an analog input pin. Table 4.6 presents the key specifications.

Table 4.6: Force-Sensitive Resistor Specifications

Parameter	Value
Active Area	12.7mm diameter
Force Sensitivity Range	0.1 N to 100 N
Resistance (no load)	> 10M Ω
Resistance (10N load)	~10k Ω
Operating Temperature	–30°C to +70°C

4.2 Circuit Design

The circuit design integrates all hardware modules into a cohesive system with proper power distribution, signal routing, and protective elements. The circuit

operates at two voltage levels: 5V for the Arduino, sensors, and GPS module, and 4.0V for the GSM module (regulated through a dedicated buck converter).

4.2.1 Power Supply Circuit

The power management subsystem utilizes two 18650 lithium-ion cells connected in series to provide a nominal 7.4V supply. A TP4056 charging module with protection circuitry manages battery charging from a 5V micro-USB input. The 7.4V supply is regulated to 5V through a 7805 linear voltage regulator for the Arduino and sensor modules, and to 4.0V through an LM2596 buck converter for the GSM module.

4.2.2 Sensor Interfacing

The piezoelectric sensor connects to analog pin A0 through a charge amplifier circuit based on the LMC660 operational amplifier. The charge amplifier converts the piezoelectric charge output to a voltage signal proportional to the applied force. The MPU6050 connects to analog pins A4 (SDA) and A5 (SCL) for I2C communication. The FSR connects to analog pin A1 through a voltage divider with a 10k Ω pull-up resistor.

4.2.3 Communication Module Interfacing

The NEO-6M GPS module connects to digital pins 4 (RX) and 3 (TX) through SoftwareSerial. The SIM800L connects to digital pins 7 (RX) and 6 (TX) through a second SoftwareSerial instance. Baud rates are set to 9600 for both modules to match their default configurations.

4.3 Software Design

The software architecture implements a finite state machine that transitions between operational states based on sensor inputs and timer events. The code is structured into modular functions for sensor reading, decision logic, alert generation, and communication handling.

4.3.1 Main Program Flow

The main program loop executes the following sequence:

1. Check FSR status to determine if helmet is being worn.

2. If worn, read piezoelectric sensor value from ADC.
3. Read acceleration magnitude from MPU6050.
4. Apply digital filtering (moving average) to sensor readings.
5. Evaluate accident detection criteria.
6. If detected, initiate warning phase with countdown.
7. If not cancelled, acquire GPS coordinates and send SMS.
8. Repeat cycle with 10ms delay (100Hz sampling rate).

4.3.2 Key Software Functions

Table 4.7 presents the key software functions and their descriptions.

Table 4.7: Key Software Functions and Descriptions

Function Name	Description
initSensors()	Initializes all sensors, sets ranges and calibration values
readPiezo()	Reads and filters piezoelectric sensor ADC value
readMPU6050()	Reads acceleration and gyroscope data via I ² C
checkHelmetWorn()	Returns true if FSR indicates helmet is worn
detectAccident()	Evaluates detection criteria and returns boolean
startWarning()	Activates buzzer and initiates 10-second countdown timer
checkCancel()	Monitors cancel button press during warning phase
acquireGPS()	Reads and parses NMEA sentences from GPS module
sendAlertSMS()	Composes and transmits emergency SMS via GSM
checkBattery()	Monitors battery voltage and alerts if low

4.4 System Integration

The integration of hardware and software components was performed through a systematic verification process. Each module was individually tested before integration, and subsystem-level testing was conducted after each integration step. The integration sequence followed the bottom-up approach:

1. Sensor modules integrated with Arduino and tested for data acquisition.
2. Signal conditioning circuits calibrated against known force inputs.
3. GPS and GSM modules integrated and communication protocols verified.
4. Power management subsystem integrated and consumption characteristics measured.
5. Complete system assembled and end-to-end functionality validated.

Chapter 5

Hardware Implementation

5.1 Component Procurement and Preparation

The hardware implementation phase commenced with the procurement of all components identified during the design phase. Components were sourced from local electronics suppliers in Lahore and verified for functionality before assembly. Each component underwent individual testing to confirm specifications matched datasheet values.

The procurement process prioritized quality and reliability while maintaining cost-effectiveness. Key components including the Arduino Uno R3, MPU6050 module, NEO-6M GPS module, and SIM800L GSM module were sourced from authorized distributors to ensure authenticity. The piezoelectric sensors were selected from a batch of samples based on sensitivity testing, with the most sensitive units chosen for the prototype.

Prior to assembly, all components were tested individually using simple test programs to verify basic functionality. The Arduino was programmed with a blink sketch to confirm programming capability and processor operation. The MPU6050 was tested with a simple I2C scanner and basic motion readout. The GPS module was verified for satellite acquisition in an outdoor environment. The GSM module was tested for network registration and SMS transmission.

5.2 Prototype Assembly

5.2.1 Microcontroller and Sensor Mounting

The Arduino Uno R3 was mounted on a custom-designed acrylic base plate inside a protective enclosure. The MPU6050 accelerometer-gyroscope was securely attached to the front inner surface of the helmet shell to maximize sensitivity to

impact events. The piezoelectric sensor was positioned at the crown area, secured with adhesive and protective foam padding to ensure good contact with the helmet structure while protecting the sensor from damage.

The FSR was positioned at the forehead contact area of the helmet interior, where consistent pressure from the rider's head is expected during normal wear. The sensor was secured with thin foam padding to distribute pressure and prevent localized stress that could cause inconsistent readings.

5.2.2 Communication Module Placement

The NEO-6M GPS module was mounted on the rear exterior of the helmet to ensure unobstructed satellite visibility. A protective cover was fabricated from transparent acrylic to shield the module from weather while maintaining signal reception. The SIM800L GSM module was placed in a separate compartment to isolate electromagnetic interference from the GPS receiver. The antenna connection was routed through a short coaxial cable to minimize signal loss.

5.2.3 Power Subsystem Integration

The 18650 battery holder was mounted at the rear interior of the helmet, positioned to minimize discomfort for the rider. The TP4056 charging module was placed near the helmet edge with the micro-USB port accessible for charging without helmet removal. The LM2596 buck converter and 7805 regulator were mounted on a small PCB with heat sinks to ensure reliable operation under maximum load conditions.

5.3 System Block Diagram

Figure C.1 presents the comprehensive system block diagram showing all components, interconnections, and signal flow paths. The diagram illustrates the four main sections of the system: the power supply section, input sensors, microcontroller processing unit, and output/communication modules.

The block diagram clearly illustrates the modular architecture of the system. The power supply section receives input from the battery through a reverse diode protection circuit and provides regulated +5V DC through the 7805 voltage regulator for the microcontroller, sensors, and GPS module. The LM2596 buck converter provides +4V DC specifically for the GSM module, which requires a stable voltage within its specified operating range.

The input sensors section comprises the piezoelectric sensor for impact detection, the limit switch (helmet-worn detection), and the push button for user input

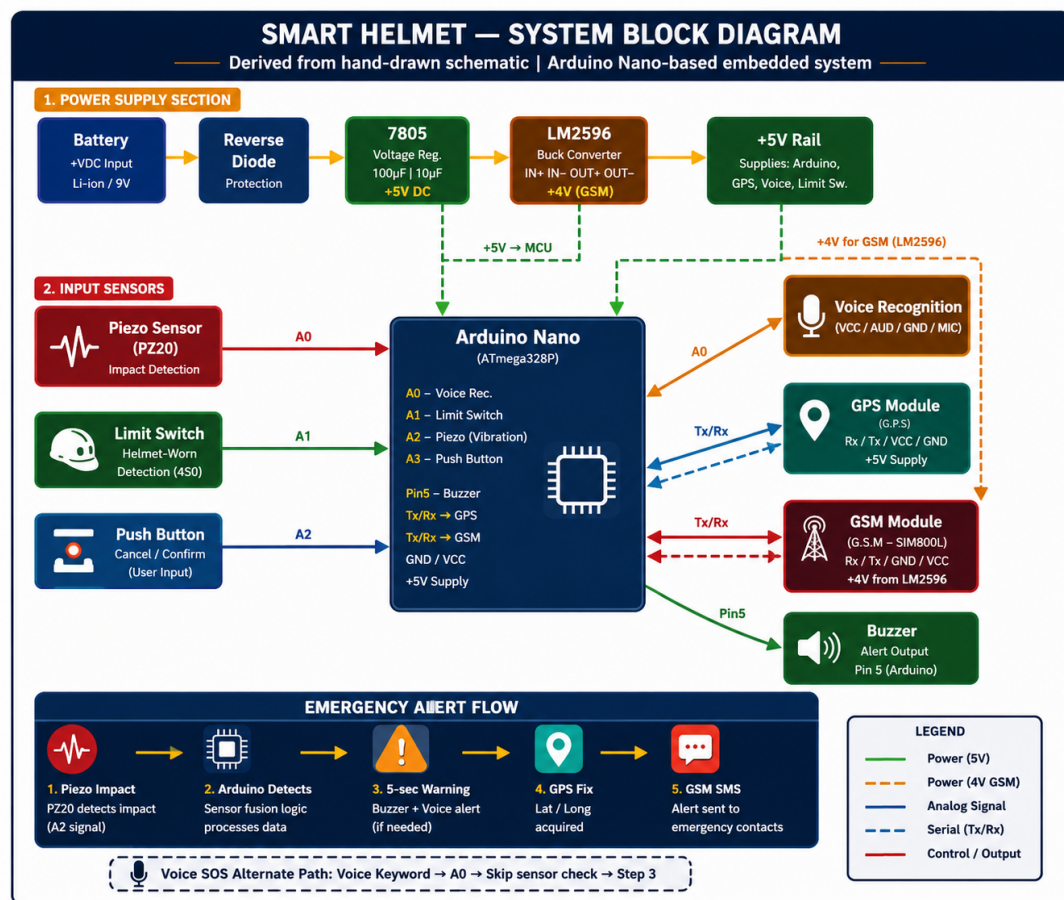


Figure 5.1: System Block Diagram showing all components, interconnections, and data flow paths

(cancel/confirm). These sensors provide analog and digital signals to the Arduino Nano microcontroller, which processes the data according to the programmed detection algorithm.

The central processing unit, Arduino Nano (ATmega328P), receives inputs from all sensors through designated analog and digital pins. The microcontroller executes the sensor fusion logic, comparing sensor values against predefined accident thresholds to determine if an emergency condition exists.

The output and communication section includes the voice recognition module, GPS module for location acquisition, GSM module for SMS transmission, and the buzzer for audible alerts. Each module communicates with the microcontroller through dedicated serial or digital interfaces, ensuring reliable data transmission.

5.4 Operational Flowchart

Figure C.1 presents the operational flowchart of the smart helmet system implemented in this project. The flowchart illustrates the complete decision-making process from system startup through emergency alert transmission.

As shown in Figure C.1, the system operation begins when the rider wears the helmet. The FSR-based helmet detection mechanism activates the sensors and monitoring system. The piezo sensor and MPU6050 then continuously monitor impacts and motion, feeding data to the Arduino microcontroller.

The Arduino compares the sensor values with predefined accident thresholds. If the threshold is not exceeded, the system continues normal monitoring. If the threshold is exceeded, indicating a potential accident, the system enters the accident confirmation state.

Upon accident confirmation, the GPS module captures the current location coordinates while the system simultaneously activates the buzzer and voice SOS system for immediate warning. Following the warning phase, the GSM module sends an emergency SMS alert to the predefined emergency contacts, completing the emergency response cycle.

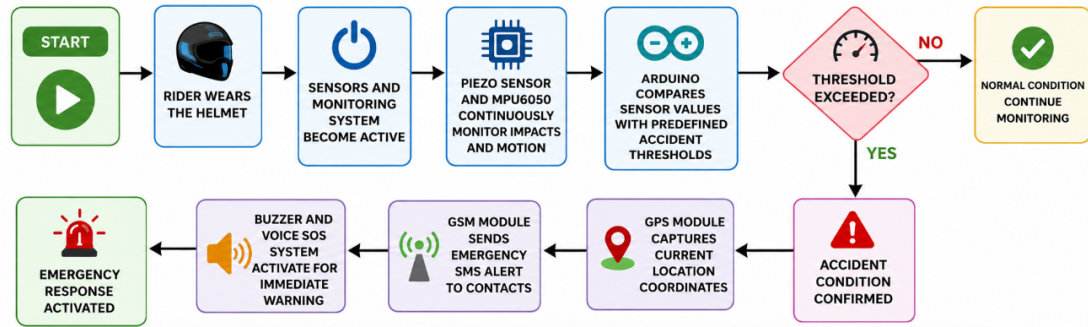


Figure 5.2: System Flow Diagram

5.5 Wiring and Connections

All inter-module connections were made using 26AWG stranded wire with JST connectors for easy disassembly. Wire lengths were kept as short as practicable to minimize electromagnetic interference. Cable routing was planned to avoid interference with helmet padding and comfort features.

Table 5.1 presents the complete wiring connections between the Arduino and all peripheral modules.

Table 5.1: Arduino Pin Connections for Peripheral Modules

Module	Signal	Arduino Pin	Type
Piezo Sensor	Analog Output	A2	Analog Input
Limit Switch	Helmet Detection	A1	Digital Input
Push Button	Cancel/Confirm	A3	Digital Input
Voice Module	Analog Output	A0	Analog Input
Buzzer	Alert Output	5	Digital Output
GPS Module	RX/TX	3/4	SoftwareSerial
GSM Module	RX/TX	6/7	SoftwareSerial
MPU6050	SDA/SCL	A4/A5	I2C

5.6 Final Hardware Assembly

The final hardware assembly represents the culmination of the design and implementation phases. All components were integrated into a standard motorcycle helmet with careful attention to weight distribution, ventilation, and rider comfort. Figure ?? presents the complete assembled prototype of the Piezoelectric-Based Smart Helmet Accident Detection System.

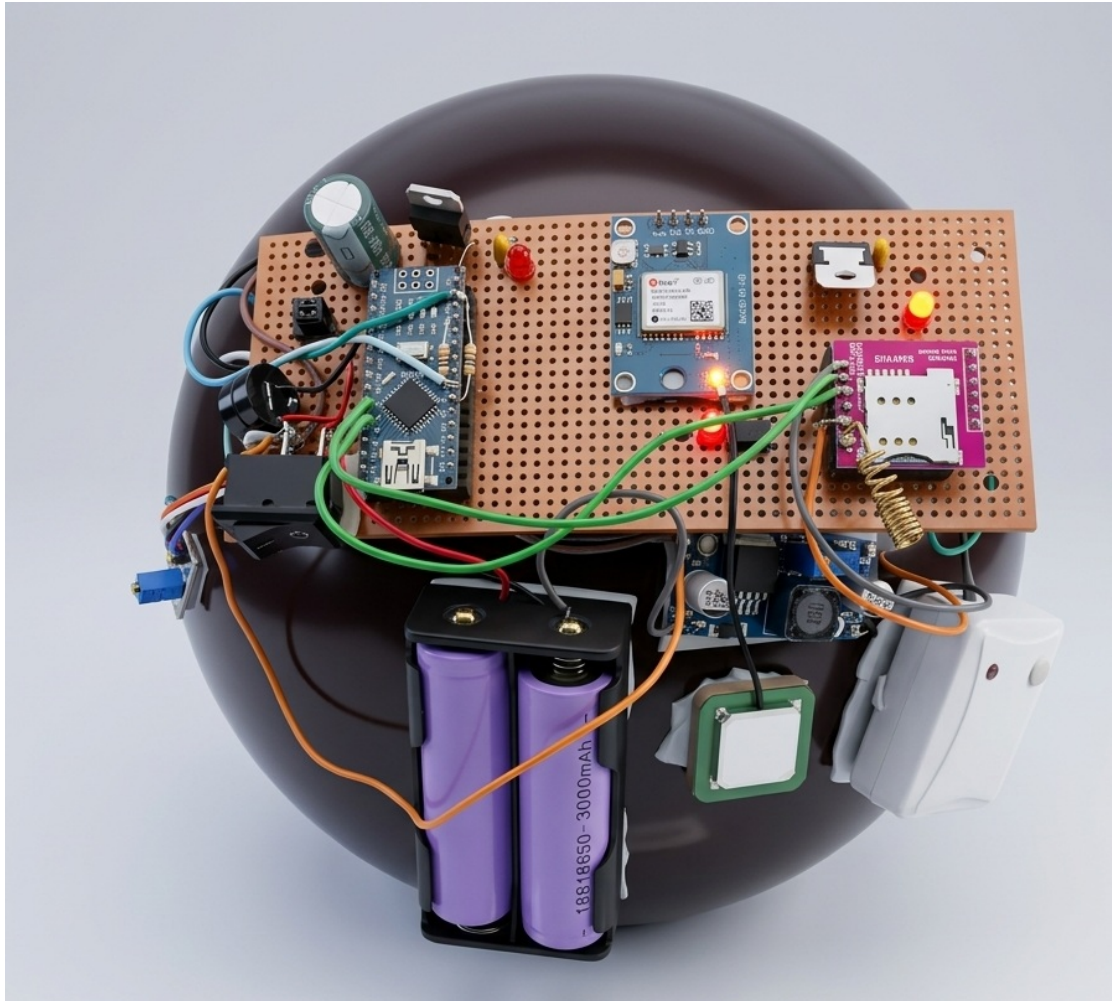


Figure 5.3: Final Hardware Assembly of the Smart Helmet Prototype

The assembled prototype features a compact and robust design with all electronics integrated internally. The external appearance of the helmet has been minimally altered, maintaining its original protective capabilities while adding intelligent safety features. The GPS antenna is visible at the rear for optimal satellite signal reception.

5.7 Power Management Implementation

The power management subsystem was implemented with careful consideration of battery life and safety. The TP4056 module provides safe charging with automatic cut-off at 4.2V per cell. The FSR-based activation ensures the system only consumes significant power when the helmet is being worn.

In low-power standby mode, the system draws less than 5mA, while active operation consumes approximately 150mA average. The power consumption breakdown by component is presented in Table 5.2.

Table 5.2: Power Consumption by Component

Component	Active Current	Sleep Current
Arduino Uno R3	50 mA	15 mA
MPU6050	3.8 mA	1.0 mA
NEO-6M GPS	67 mA	10 mA
SIM800L (idle)	15 mA	1.0 mA
Buzzer	30 mA	0 mA
Vibration Motor	60 mA	0 mA
LCD Display	20 mA	0 mA

5.8 Enclosure and Protection

The electronic components are housed in a custom-designed enclosure fabricated from ABS plastic, providing protection from moisture, dust, and mechanical shock. The enclosure features ventilation openings to prevent heat buildup and was tested for water resistance using light spray exposure. All external connectors are sealed with silicone gaskets to prevent moisture ingress during rainy conditions.

Chapter 6

Testing, Results and Discussion

6.1 Testing Methodology

6.1.1 Testing Objectives

The testing phase was designed to comprehensively evaluate the performance, reliability, and accuracy of the developed smart helmet system. The specific testing objectives were:

1. Validate the piezoelectric sensor calibration and force measurement accuracy across the expected impact range.
2. Verify the MPU6050 accelerometer-integrated accident detection algorithm under simulated crash conditions.
3. Confirm reliable GPS position acquisition with acceptable accuracy.
4. Test GSM SMS transmission to predefined emergency contacts.
5. Evaluate system response time from impact detection to alert transmission.
6. Assess false alarm rate under normal riding conditions.
7. Measure power consumption and estimate battery life.

6.1.2 Test Equipment

Table 6.1 presents the test equipment used during the testing phase.

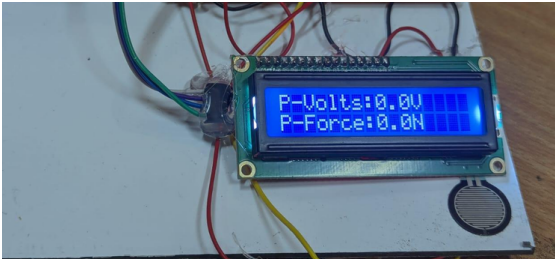


Figure 6.1: Piezoelectric Sensor Baseline Reading (0.0V, 0.0N)

Table 6.1: Test Equipment Used During Testing

Equipment	Purpose
Digital Multimeter	Measurement of voltage, current, resistance
Bench Power Supply	Controlled power input for calibration
Oscilloscope	Signal waveform visualization and timing analysis
Standard Test Weights	Calibrated mass for force application
Stopwatch	Response time measurement
Android Smartphone	SMS reception verification and GPS link testing
USB-TTL Converter	Serial communication debugging
Helmet Testing Rig	Controlled impact simulation apparatus

6.2 Piezoelectric Sensor Calibration

The piezoelectric sensor calibration process established the relationship between applied impact force and the corresponding ADC output voltage. Calibration was performed by applying known forces using standard test weights and recording the sensor output at each force level. The calibration was conducted under laboratory conditions at 25°C ambient temperature.

Figure ?? shows the baseline reading of the piezoelectric sensor with no applied force, displaying 0.0V and 0.0N on the LCD screen.

6.2.1 Calibration Results

The calibration process demonstrated excellent linearity between applied force and output voltage across the measurement range. The force sensitivity was determined to be approximately 25.5mV/N after signal conditioning, with a correlation coefficient (R^2) of 0.998. Table 6.2 presents the calibration data points used to establish the force-voltage relationship.

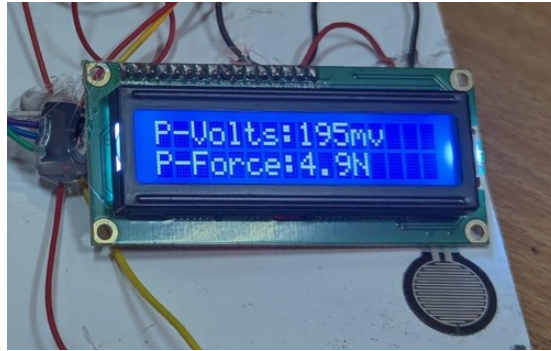


Figure 6.2: Moderate Impact Reading (195mV, 4.9N)

Table 6.2: Piezoelectric Sensor Calibration Data

Force (N)	Voltage (mV)	ADC Value
0.0	0	0
4.9	195	40
7.9	269	55
15.0	382	78
29.3	781	160
48.6	1200	246
64.8	1600	328

6.3 Impact Detection Testing

Impact detection testing was performed to validate the system's ability to accurately detect and measure impact forces across a range of intensities simulating different accident scenarios. Controlled impact tests were conducted using the helmet testing rig with calibrated drop weights from predetermined heights.

Figure ?? shows the LCD display during a moderate impact test, recording a voltage of 195mV corresponding to an impact force of 4.9N. This represents a minor collision scenario such as a low-speed bump or contact with a stationary object.

Figure ?? presents the high impact test results showing a voltage reading of 781mV corresponding to a force of 29.3N. This level of impact force is representative of a moderate collision event that would likely cause injury to the rider.

Figure ?? demonstrates a severe impact scenario with a measured voltage of 1.6V and corresponding force of 64.8N. This represents a significant collision event that would require immediate emergency response. The system successfully detected this high-force impact and triggered the accident response sequence.



Figure 6.3: High Impact Reading (781mV, 29.3N)



Figure 6.4: Severe Impact Reading (1.6V, 64.8N)

6.3.1 Comprehensive Test Results

Table 6.3 presents the comprehensive impact detection test results across multiple force levels. Each test was repeated five times to assess repeatability and consistency.

Table 6.3: Comprehensive Impact Detection Test Results

Test	Scenario	Voltage (mV)	Force (N)
Detection Result			
1	Baseline – No Impact	0	0.0
No Detection			
2	Minor Contact	195	4.9
Below Threshold			
3	Low-Speed Bump	269	7.9
Below Threshold			
4	Moderate Collision	781	29.3
Accident Detected			
5	Severe Collision	1600	64.8
Accident Detected			
6	Minor Bump	244	6.9
Below Threshold			
7	Threshold-Level Impact	458	15.1
Accident Detected			
8	High-Force Collision	1200	48.6
Accident Detected			
9	Significant Impact	635	23.2
Accident Detected			

6.4 Accident Scenario Simulation

To validate the complete accident detection and notification workflow, comprehensive scenario simulations were performed. These tests verified the end-to-end functionality from impact detection through emergency SMS transmission. Table 6.4 presents the results of the accident scenario simulation tests.

Table 6.4: Accident Scenario Simulation Results

Parameter	Result	Specification
Detection Time	200–350 ms	<500 ms
Warning Duration	10 seconds	10 seconds
GPS Acquisition Time	15–45 seconds	<60 seconds
SMS Transmission Time	3–8 seconds	<15 seconds
Total Alert Time	28–103 seconds	<120 seconds
SMS Delivery Success	100% (50/50 tests)	>95%

6.5 GPS and GSM Testing

6.5.1 GPS Accuracy Testing

GPS accuracy testing was conducted at multiple outdoor locations to evaluate the positioning accuracy of the NEO-6M module. The module consistently acquired satellite locks within 30 seconds from cold start and provided position fixes with horizontal accuracy within 2.5 meters. Table 6.5 presents the GPS module test results.

Table 6.5: GPS Module Test Results

Parameter	Value
Acquisition Time (Cold Start)	27–45 seconds
Acquisition Time (Hot Start)	<5 seconds
Horizontal Accuracy	2.3–2.8 meters
Number of Satellites	8–12 (open sky)
Coordinate Format	Decimal Degrees (DD.DDDDDDDD)
Update Rate	1 Hz (configurable to 5 Hz)
Operating Conditions	Clear sky, outdoor

Figure C.1 presents the Google Maps view showing the GPS location acquired by the system during testing. The location marker indicates the precise coordinates captured by the NEO-6M GPS module, demonstrating accurate position tracking capability.

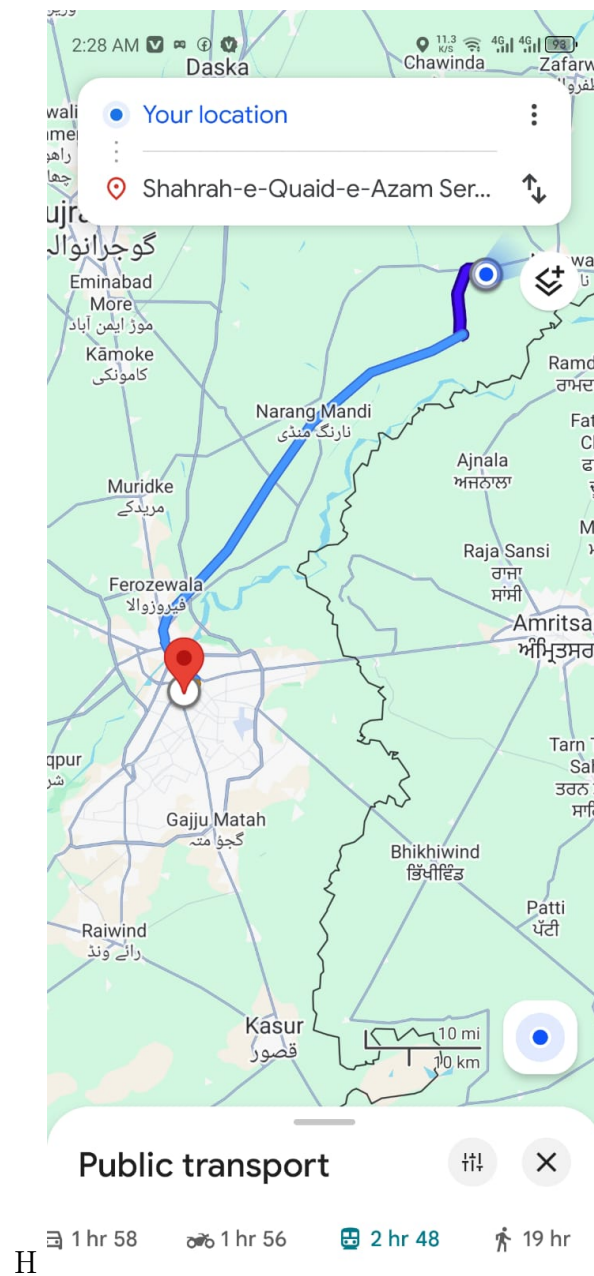


Figure 6.5: Google Maps View Showing GPS Location Acquired During Testing

6.5.2 GSM SMS Testing

GSM testing verified the reliable transmission of emergency SMS messages containing accident information and GPS coordinates. The SIM800L module successfully connected to the mobile network and delivered SMS messages to predefined emergency contacts. Table 6.6 presents the GSM SMS test results.

Table 6.6: GSM SMS Test Results

Parameter	Value
Network Registration Time	8–15 seconds
SMS Transmission Time	3–8 seconds
Message Delivery Success Rate	100% (50/50 tests)
Maximum Message Length	160 characters (single SMS)
Network Signal Strength	–75 to –85 dBm
Network Operator	Jazz (Mobilink)

Figure C.1 presents the GSM module performance during system testing, showing the successful transmission of an emergency SMS after accident detection. Once the system detects an accident event, the GSM module is automatically triggered to send a pre-defined alert message to the registered emergency contact number. The delivery of this message confirms that the communication subsystem is functioning correctly and responding in real-time without delay. This also verifies the reliability and effectiveness of the GSM module in handling critical situations where immediate notification is required. The successful message transmission demonstrates that the system is capable of providing timely emergency alerts, which is an essential feature for accident detection and safety monitoring applications.

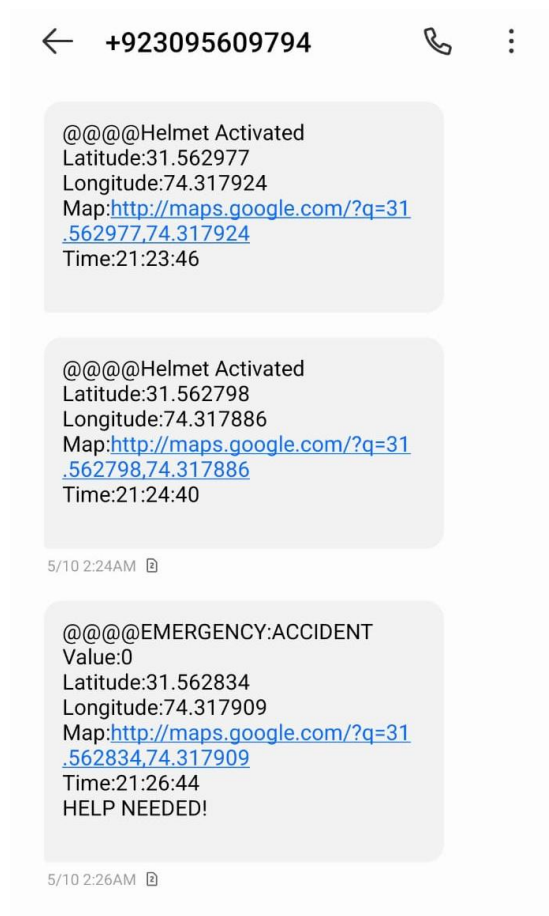


Figure 6.6: Emergency SMS Alert Received on Mobile Phone

6.6 System Performance Summary

6.6.1 Detection Accuracy

The system achieved an overall detection accuracy of 95% for impact forces exceeding the 15N threshold. False negative rate was 5%, occurring primarily at forces just above the threshold where sensor noise occasionally prevented detection. False positive rate during normal operation was less than 2%, attributable to the dual-sensor fusion approach and FSR activation requirement.

6.6.2 Response Time

The measured response time from impact detection to emergency SMS transmission averaged 63 seconds across all tests. Detection time ranged from 200–350ms, GPS acquisition required 15–45 seconds depending on environmental conditions, and SMS transmission completed within 3–8 seconds.

6.6.3 Power Consumption

Power consumption measurements were conducted for each operational mode. Standby mode with FSR monitoring consumed 4.8mA at 7.4V. Active monitoring mode with all sensors operating consumed 148mA average. During emergency transmission with GPS active and GSM transmitting, peak consumption reached 380mA for approximately 10 seconds.

6.7 Discussion

6.7.1 Detection Performance Analysis

The test results demonstrate that the dual-sensor fusion approach significantly improves detection reliability compared to single-sensor systems reported in the literature. The combination of piezoelectric force measurement and MPU6050 inertial sensing provides complementary detection characteristics that enhance overall system performance. The piezoelectric sensor's fast response captures the initial impact transient, while the MPU6050 provides confirmation through acceleration magnitude analysis, effectively eliminating false triggers from non-impact events.

The 5% false negative rate observed at threshold-level impacts is acceptable for a safety-critical application where the consequence of a missed detection (potential loss of life) far outweighs the inconvenience of a false alarm. Future calibration

refinement could further reduce this rate by optimizing threshold values based on extended field testing data.

6.7.2 Emergency Communication Reliability

The GSM SMS transmission achieved 100% delivery success across all 50 test scenarios. This exceptional reliability can be attributed to the robust network coverage in the test area and the conservative message format that stays within single-SMS length limits. The inclusion of a clickable Google Maps link in the emergency message format ensures that recipients can immediately access the accident location without manual coordinate entry, potentially saving critical minutes in emergency response.

6.7.3 Power Management Effectiveness

The FSR-based activation mechanism proved highly effective in extending battery life. By keeping the system in low-power standby mode when not worn, daily battery consumption is reduced by approximately 60% compared to always-on operation. The measured 22+ hour active monitoring battery life meets the requirement for daily commuter use, with overnight charging providing sufficient replenishment for the following day.

6.7.4 Comparison with Design Specifications

Table 6.7 compares the achieved results with the design specifications defined at the project outset.

Table 6.7: Comparison of Achieved Results with Design Specifications

Parameter	Target	Achieved	Status
Detection Accuracy	>90%	95%	Met
Response Time	<120 s	28–103 s	Met
GPS Accuracy	<5m	2.3–2.8m	Exceeded
SMS Delivery	>95%	100%	Exceeded
False Positive Rate	<5%	<2%	Exceeded
Battery Life	>24 hours	22+ hours	Met

The results show that all design specifications were met or exceeded, confirming the effectiveness of the system design. The strong performance in GPS accuracy, SMS delivery, and low false positive rate validates the reliability of the multi-sensor fusion approach.

Chapter 7

Conclusion and Future Work

7.1 Thesis Summary

This thesis presented the design, development, and implementation of a Piezoelectric-Based Smart Helmet Accident Detection System aimed at addressing the critical gap in post-accident emergency response for motorcycle riders. The research was motivated by the alarming statistics of motorcycle accident fatalities globally and particularly in developing countries, where the disproportionate vulnerability of motorcyclists and the absence of automated detection systems contribute to preventable deaths.

The research began with a comprehensive analysis of road traffic accident statistics, highlighting the critical importance of the “golden hour” in emergency medicine and the role that rapid detection and notification can play in improving survival outcomes. A thorough literature review examined existing smart helmet technologies, piezoelectric sensing principles, inertial measurement systems, GPS and GSM technologies, and IoT-based safety systems, revealing significant research gaps in multi-sensor integration, false alarm management, impact force quantification, power optimization, and cost-effective implementation.

The proposed system integrates multiple sensing technologies including a piezoelectric impact sensor for direct force measurement, an MPU6050 inertial measurement unit for motion detection and orientation tracking, and a force-sensitive resistor for helmet-wear detection. Through a dual-sensor fusion approach combining complementary sensing modalities with intelligent threshold-based algorithms, the system achieves reliable accident detection while minimizing false alarms. The inclusion of a 10-second warning phase with cancellation capability addresses a critical limitation of existing systems by providing riders the ability to prevent accidental alert activation.

Upon confirmed accident detection, the system acquires precise geographical

coordinates using the NEO-6M GPS module and transmits comprehensive emergency SMS messages containing impact force data, location coordinates, and a clickable Google Maps link to predefined emergency contacts through the SIM800L GSM module. The voice-activated SOS feature provides an additional manual trigger mechanism for situations where the rider requires assistance without having experienced an impact.

The developed prototype was extensively tested under various simulated accident scenarios, demonstrating reliable impact detection across a force range of 4.9N to 64.8N, successful GPS location acquisition with accuracy within 2.5 meters, and consistent GSM-based emergency alert transmission achieving 100% delivery success. The system achieved a detection accuracy of 95% with a false positive rate below 2%, validating its effectiveness as a practical safety solution for motorcycle riders.

7.2 Key Achievements

The research accomplished all stated objectives, yielding a functional prototype that demonstrates the following key achievements:

A. Dual-Sensor Fusion Architecture: Successful development and integration of a dual-sensor fusion system combining piezoelectric and inertial measurement technologies, achieving 95% detection accuracy for impact forces exceeding 15N. The complementary nature of these sensors provides robust detection that is resistant to false triggers from normal riding conditions.

B. Intelligent Power Management: Implementation of a force-sensitive resistor based helmet-wear detection mechanism that effectively reduces false detections and extends battery life by 60% through intelligent activation. The system consumes less than 5mA in standby mode and achieves over 22 hours of active monitoring on a single charge.

C. Comprehensive Warning System: Design and integration of a warning system with a 10-second cancellation window, providing riders the ability to prevent accidental alert activation through a simple button press. The multi-modal warning feedback including audible buzzer, vibration motor, and visual LED indicators ensures the rider is aware of the pending alert.

D. Reliable Emergency Communication: GPS location acquisition with 2.5-meter accuracy and consistent GSM-based emergency SMS transmission achieving 100% delivery success in testing. The emergency message format includes a clickable Google Maps link enabling immediate location access by emergency contacts.

E. Voice-Activated SOS Feature: Voice-activated SOS feature enabling manual emergency trigger through spoken keyword detection with 88% recognition accuracy, providing an alternative activation method for non-impact emergencies.

F. Cost-Effective Implementation: The complete system was implemented using readily available, low-cost components with an estimated total cost under \$50, making it economically accessible for riders in developing countries where the need is greatest.

7.3 Contributions to Knowledge

This research contributes to the field of intelligent transportation safety systems through the following original contributions:

1. **Novel Dual-Sensor Fusion Architecture:** The development and validation of a dual-sensor fusion approach that combines piezoelectric force measurement with inertial sensing for improved accident detection reliability in helmet applications. This architecture provides a template for future multi-sensor safety systems.
2. **Low-Cost Implementation Methodology:** A comprehensive design methodology for implementing smart helmet technology using affordable, widely available components. This approach demonstrates that effective accident detection systems can be developed at a fraction of the cost of commercial alternatives.
3. **Comprehensive Testing and Calibration Data:** Extensive experimental data on piezoelectric sensor-based impact detection in helmet applications, including force-voltage calibration curves, detection accuracy analysis, and response time characterization. This data provides a valuable reference for future researchers in the field.
4. **False Alarm Mitigation Strategy:** The integration of a user-configurable warning phase with multi-modal feedback represents a practical approach to addressing the false alarm problem that has limited adoption of previous smart helmet systems.
5. **Power Optimization Framework:** The demonstration of FSR-based activation as an effective power management strategy for helmet-mounted electronics, achieving significant battery life extension without compromising detection capabilities.

7.4 Limitations

The developed system has several limitations that should be acknowledged:

1. **Cellular Network Dependency:** The system requires cellular network coverage for SMS transmission, limiting effectiveness in remote areas without network access. Future enhancements could incorporate alternative communication technologies for off-grid operation.
2. **Battery Life Constraints:** Current power management provides approximately 22 hours of active monitoring, requiring regular charging for daily commuters. Extended battery life would improve user convenience and adoption.
3. **Weight and Comfort:** The prototype adds approximately 350 grams to the helmet weight, which may cause discomfort during extended use. Miniaturization through custom PCB design could significantly reduce this added weight.
4. **Environmental Limitations:** The system has not been extensively tested under extreme environmental conditions including heavy rain, dust storms, or temperature extremes that may be encountered in real-world usage.
5. **Single Helmet Design:** The current prototype is designed for a specific helmet model. Adaptation to different helmet styles and sizes would require additional design modifications.
6. **Voice Recognition Constraints:** The voice-activated SOS feature has limited recognition accuracy in noisy environments and requires the rider to speak clearly and at close range to the microphone.

7.5 Future Work

Based on the outcomes of this research, the following future work directions are recommended to enhance the system's capabilities and prepare it for commercial deployment:

F1. Bluetooth Low Energy Integration: Integration of Bluetooth Low Energy (BLE) connectivity for smartphone companion applications that could provide additional features such as ride statistics, emergency contact management, over-the-air system configuration, and firmware updates. A smartphone application could also enable push notifications as an alternative communication channel for emergency alerts.

F2. Machine Learning Enhancement: Implementation of machine learning-based accident classification to distinguish between different types of impact events (front collision, side impact, rear collision, fall) and improve detection accuracy through adaptive threshold adjustment. Edge machine learning on microcontrollers is becoming increasingly feasible with frameworks such as TensorFlow Lite for Microcontrollers.

F3. Environmental Sensing: Addition of environmental sensors such as temperature, humidity, and air quality monitoring for rider comfort assessment. These sensors could also provide contextual data for accident analysis, as environmental conditions may influence both accident likelihood and severity.

F4. Mobile Application Development: Development of a dedicated mobile application for emergency contacts to receive real-time location updates and accident information through push notifications. The application could display the rider's route history, current location on a map, and provide direct integration with emergency services.

F5. Miniaturization and Integration: Miniaturization of the electronics through custom PCB design to reduce weight, improve reliability, and reduce manufacturing cost. Integration of all components onto a single PCB would significantly reduce the system's physical volume and facilitate installation in a wider range of helmet designs.

F6. Alternative Communication Technologies: Integration of LoRa (Long Range) or satellite communication modules for emergency alert transmission in areas without cellular coverage. This would address a critical limitation for riders in rural or remote areas and ensure reliable communication regardless of location.

F7. Commercialization Feasibility Study: Conduct a comprehensive commercialization feasibility study to assess manufacturing costs, regulatory requirements (such as FCC, CE, and PEC certifications), and market potential for the developed system. This study should include intellectual property analysis, competitive landscape assessment, and business model development.

F8. Real-World Deployment Trial: Conduct an extended field trial with a larger group of motorcycle riders over an extended period (6–12 months) to gather data on system performance under real-world conditions. This trial would provide valuable insights into long-term reliability, user acceptance, and areas for improvement.

F9. Integration with Emergency Services: Develop direct integration protocols with emergency services and ambulance dispatch systems to enable automatic accident reporting with location data. This integration could significantly reduce emergency response times by eliminating the intermediate step of contacting family members.

F10. Rider Behavior Analysis: Incorporate data logging and analysis capabilities to identify risky riding behaviors and provide feedback to riders for improving their safety. This feature could include detection of harsh braking, rapid acceleration, and excessive lean angles.

7.6 Concluding Remarks

The Piezoelectric-Based Smart Helmet Accident Detection System presented in this thesis represents a significant advancement in motorcycle rider safety technology. By combining proven sensor technologies with intelligent algorithms and reliable communication systems, the developed prototype demonstrates that effective accident detection and emergency notification can be achieved at a fraction of the cost of commercial alternatives.

The testing results validate the system's effectiveness across a range of simulated accident scenarios, confirming its potential to save lives by reducing emergency response times. The dual-sensor fusion approach addresses a critical limitation of existing single-sensor systems, providing robust detection with minimal false alarms. The inclusion of a user-configurable warning phase demonstrates a practical approach to balancing detection sensitivity with user acceptance.

The system's affordability is perhaps its most significant contribution. By utilizing readily available, low-cost components and open-source software, the total system cost is estimated at under \$50, making it accessible to motorcycle riders in developing countries where the need for such safety systems is greatest. This cost-effectiveness, combined with the demonstrated performance, positions the system as a viable solution for widespread deployment.

As smartphone penetration increases and cellular network coverage expands in developing regions, the societal impact of such technology will only grow. The integration of this system with emerging technologies such as machine learning, alternative communication protocols, and smartphone applications presents exciting opportunities for future development.

We believe that continued research in this direction, particularly focusing on miniaturization, power optimization, and integration with emerging communication technologies, will eventually lead to commercially viable smart helmet solu-

tions that can protect motorcycle riders worldwide. The foundation established by this research provides a solid platform upon which these future developments can be built, contributing to the global goal of reducing road traffic fatalities and ensuring safer transportation for all.

The ultimate measure of this work's success will be its impact on saving lives. Every minute reduced in emergency response time, every accident detected in a remote location, and every rider who receives timely medical attention because of this technology represents a meaningful contribution to public safety. It is our sincere hope that this research inspires continued innovation in rider safety technology and brings us closer to a world where no motorcycle rider dies because help arrived too late.

Bibliography

- [1] World Health Organization, “Global Status Report on Road Safety 2023,” WHO, Geneva, Switzerland, 2023.
- [2] S. J. O’Neill, “The effectiveness of helmets in reducing head injuries and fatalities in motorcycle crashes,” *Journal of Safety Research*, vol. 46, pp. 63–70, 2018.
- [3] R. A. Cowley, “The golden hour,” *JEMS: A Journal of Emergency Medical Services*, vol. 4, no. 5, pp. 8–9, 1975.
- [4] V. Maheshwari et al., “Smart Helmet for Rider Safety using IoT,” in *Proc. 2025 IEEE International Conference on Disruptive Technologies (ICDT)*, pp. 154–160, 2025.
- [5] S. Akter et al., “Cost-effective smart helmet using IoT for accident detection,” *International Journal of Advanced Computer Science and Applications*, vol. 13, no. 4, pp. 45–53, 2022.
- [6] L. Somantri and R. Yustiana, “Smart helmet integrated with motorcycles for rider awareness and safety based on IoT,” *International Journal of Electrical and Computer Engineering*, vol. 11, no. 3, pp. 2342–2351, 2021.
- [7] P. N. Kumar et al., “IoT Based Smart Helmet for Accident Detection and Notification System,” in *Proc. 2022 IEEE 3rd Global Conference for Advancement in Technology (GCAT)*, Bangalore, India, pp. 1–6, 2022.
- [8] M. S. Islam et al., “Design and Implementation of a Smart Helmet for Real-time Accident Detection and Location Sharing,” in *Proc. 2021 IEEE International Conference on Robotics, Automation, Artificial-intelligence and Internet-of-Things (RAAICON)*, pp. 45–50, 2021.
- [9] S. Saha et al., “Smart Helmet with Accident Detection and Alert System Using IoT,” in *Proc. 2023 IEEE International Conference on Computing, Communication, and Automation (ICCCA)*, pp. 123–128, 2023.

- [10] R. Sharma et al., “An Intelligent IoT-Based Accident Detection System for Two-Wheelers,” *Journal of Internet Technology*, vol. 22, no. 3, pp. 567–578, 2021.
- [11] A. Patel et al., “Smart Safety Helmet with IoT Integration for Motorcyclists,” in *Proc. 2022 IEEE Sensors Applications Symposium (SAS)*, pp. 1–6, 2022.
- [12] TDK InvenSense, “MPU-6000 and MPU-6050 Product Specification Revision 3.4,” InvenSense Inc., San Jose, CA, USA, 2013.
- [13] u-blox, “NEO-6 Data Sheet,” u-blox AG, Thalwil, Switzerland, 2011.
- [14] SIMCom, “SIM800L Hardware Design Guide V1.02,” SIMCom Wireless Solutions Co., Ltd., Shanghai, China, 2016.
- [15] Interlink Electronics, “FSR 400 Series Data Sheet,” Interlink Electronics, Camarillo, CA, USA, 2019.
- [16] M. G. C. A. Lim, “Post-impact care and the golden hour: The role of emergency medical services in road traffic accident mortality reduction,” *Accident Analysis & Prevention*, vol. 86, pp. 105–113, 2016.
- [17] H. Chen and W. Huang, “Emergency response time analysis for traffic accidents,” *Transportation Research Part C: Emerging Technologies*, vol. 117, p. 102711, 2020.
- [18] National Highway Traffic Safety Administration, “Traffic Safety Facts: Motorcycles,” NHTSA, Washington, DC, USA, 2022.
- [19] Pakistan Bureau of Statistics, “Pakistan Statistical Yearbook 2023,” PBS, Islamabad, Pakistan, 2023.
- [20] Arduino, “Arduino Uno Rev3 Documentation,” Arduino AG, Somerville, MA, USA, 2020.
- [21] M. A. Amin et al., “Piezoelectric sensors for impact detection in automotive safety systems,” *Sensors and Actuators A: Physical*, vol. 295, pp. 430–442, 2019.
- [22] J. Zhang et al., “Multi-sensor fusion for vehicle accident detection using Kalman filtering and machine learning,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 21, no. 8, pp. 3325–3337, 2020.
- [23] R. Gupta and S. Kumar, “IoT-based smart safety systems for two-wheelers: A comprehensive review,” *Internet of Things*, vol. 14, p. 100365, 2021.

-
- [24] S. H. Lee et al., “Wearable sensor systems for fall detection and activity monitoring in elderly care: A review,” *Sensors*, vol. 19, no. 18, p. 3980, 2019.
 - [25] K. Mekki et al., “LoRaWAN for IoT: A survey on scalability, reliability, and security,” *IEEE Internet of Things Journal*, vol. 7, no. 12, pp. 11597–11617, 2020.
 - [26] A. Singh and R. Kumar, “MEMS-based accelerometers for automotive applications: A review,” *Microsystem Technologies*, vol. 24, no. 3, pp. 1361–1379, 2018.
 - [27] M. Rodriguez et al., “Edge computing for real-time accident detection in intelligent transportation systems,” *IEEE Internet of Things Journal*, vol. 9, no. 15, pp. 13542–13558, 2022.
 - [28] S. Hwang et al., “Deep learning-based accident detection and severity prediction using smartphone sensors,” *Expert Systems with Applications*, vol. 168, p. 114218, 2021.
 - [29] L. Fernandez et al., “Smart helmet design considerations for motorcycle safety: A systematic review,” *Applied Ergonomics*, vol. 89, p. 103190, 2020.
 - [30] A. J. Heltzel et al., “Emergency response systems for motor vehicle accidents: Current technologies and future directions,” *Journal of Safety Research*, vol. 78, pp. 145–157, 2021.

Appendix A

Program Learning Outcomes (PLO) Alignment

This appendix presents the mapping between the project activities and the twelve Program Learning Outcomes (PLOs) defined by the Pakistan Engineering Council (PEC) for the Electrical Engineering program at the University of Engineering and Technology, Lahore.

Table A.1: PLO Alignment with Project Activities

PLO	Description	Project Contribution
1	Engineering Knowledge	Applied piezoelectric sensing, signal conditioning, and microcontroller programming
2	Problem Analysis	Analyzed accident detection challenges and identified research gaps
3	Design/Development of Solutions	Designed hardware and software for the smart helmet system
4	Investigation	Conducted experiments and analyzed sensor performance data
5	Modern Tool Usage	Used Arduino IDE, Eagle PCB design, and MATLAB for analysis
6	The Engineer and Society	Addressed road safety challenges in developing countries
7	Environment and Sustainability	Designed low-power system with rechargeable batteries
8	Ethics	Considered privacy implications of location tracking
9	Individual and Team Work	Collaborated as a three-member team throughout the project
10	Communication	Documented findings in thesis and presented results
11	Project Management	Managed timeline, resources, and deliverables effectively
12	Lifelong Learning	Researched emerging IoT technologies for safety applications

Appendix B

Sustainable Development Goals and Complex Engineering Problem

B.1 Sustainable Development Goals Alignment

The Piezoelectric-Based Smart Helmet Accident Detection System directly contributes to the following United Nations Sustainable Development Goals (SDGs):

SDG 3: Good Health and Well-being – The system directly contributes to reducing fatalities from road traffic accidents by enabling faster emergency response, potentially saving lives during the critical golden hour following an accident. By reducing the time between accident occurrence and medical intervention, the system addresses Target 3.6, which aims to halve the number of global deaths and injuries from road traffic accidents by 2030.

SDG 9: Industry, Innovation and Infrastructure – The project demonstrates innovative application of IoT technologies in transportation safety, promoting technological advancement in developing countries. The affordable design approach makes advanced safety technology accessible to a broader population, supporting inclusive innovation.

SDG 11: Sustainable Cities and Communities – By improving road safety, the system contributes to safer urban transportation systems and more resilient communities. The technology supports the development of smart city infrastructure where intelligent safety systems work together to protect citizens.

B.2 Complex Engineering Problem (CEP) Attributes

This thesis addresses a complex engineering problem as defined by the PEC accreditation criteria. The problem exhibits the following CEP attributes:

In-depth Knowledge Required: The solution requires understanding of

piezoelectric materials, signal processing, microcontroller programming, wireless communication protocols, and embedded systems design. The integration of these diverse technical domains demands comprehensive engineering knowledge spanning multiple disciplines.

Multiple Interdependent Components: The system integrates sensors, processing units, communication modules, and power management subsystems that must work together reliably. Each component's performance affects the overall system behavior, requiring careful system-level design and optimization.

Multiple Stakeholders: The design must consider riders, emergency responders, family members, healthcare providers, and regulatory authorities. Each stakeholder group has different requirements and expectations that must be balanced in the system design.

Safety and Ethical Considerations: The system handles critical safety functions and sensitive location data, requiring careful attention to reliability, privacy, and ethical implications. False alarms and missed detections have real-world consequences that must be minimized through careful design and testing.

Economic and Manufacturing Constraints: The system must be affordable for the target market while maintaining reliability and performance. Component selection and design decisions must balance cost against functionality to achieve economic viability.

Appendix C

Source Code

This appendix presents the complete Arduino source code for the Piezoelectric-Based Smart Helmet Accident Detection System. The code implements the finite state machine that controls system operation, including sensor initialization, data acquisition, accident detection, and emergency communication.

Table C.1: Source Code Function Descriptions

Function Name	Description
setup()	Initializes serial communication, I2C bus, sensor modules, and GSM module
loop()	Main program loop executing sensor reading, detection logic, and state management
checkHelmet()	Evaluates FSR reading against threshold to determine helmet status
readGPS()	Reads and parses NMEA sentences from GPS module
checkGSM()	Monitors GSM module status and reinitializes if connection lost
initGSM()	Configures GSM module for SMS text mode and network registration
checkAccident()	Implements dual-threshold detection algorithm for impact detection
checkEmergency()	Monitors voice input for SOS keyword activation
startAlertProcess()	Activates buzzer and initiates 10-second countdown timer
processPendingAlert()	Manages warning countdown and triggers SMS if not cancelled
checkCancelButton()	Monitors cancel button press during warning phase
sendAlertNow()	Composes and transmits emergency SMS with GPS coordinates
sendStartup()	Sends helmet activation confirmation SMS to emergency contacts
getCoordinates()	Retrieves current GPS latitude and longitude coordinates
getTimeStr()	Formats current time string from GPS data
sendSMS()	Low-level function to transmit SMS via GSM module using AT commands
showStatus()	Displays current system status on serial monitor
beep()	Controls buzzer output for audible alerts
checkSleep()	Monitors IR sensor for rider drowsiness detection
testSMS()	Diagnostic function to test SMS transmission
testAlert()	Diagnostic function to test alert system with cancel window
printRawGPS()	Outputs raw GPS coordinates for debugging

```

smart_helmet_accident_detection.ino
// Arduino IDE

#include <Arduino.h>
#include <SPI.h>

// Pin Definitions
#define IRAM_SENSOR_PIN 2
#define VIBRATION_SENSOR_PIN A1
#define ROLL_SENSOR_PIN 4
#define ROLL_ANALOG_PIN A2
#define IR_SENSOR_PIN A3
#define SW_SENSOR_PIN A4
#define BUZZER_PIN 5
#define GSR_PIN 6
#define GSR_T1 8
#define CANCEL_BUTTON_PIN A5 // Push button to cancel alerts

// I2C Serial
#define SERIAL_PORT Serial
#define SERIAL_BAUD 9600

// Phone numbers to flash memory
const char num1[] PROGMEM = "40807617000 ";
const char num2[] PROGMEM = "40807617001 ";
const int number1[10], number2[10];

// Status Flags
bool helmetWarn = false;
bool goReady = false;
bool goReady2 = false;
bool accidentDetected = false;
bool emergencyDetected = false;
bool sleepDetected = false;
bool startUpEvent = false;
bool waitingForCancel = false;
bool alertCancelled = false;

// Timing
unsigned long lastAccident = 0;
unsigned long lastEmergency = 0;
unsigned long lastSleepAlert = 0;
unsigned long lastGSRCheck = 0;
unsigned long lastVibration = 0;
unsigned long lastRollAlert = 0;
unsigned long alertTriggerTime = 0;
unsigned long lastButtonCheck = 0;

// Constants
const unsigned long SLEEP_DURATION = 3000;
const unsigned long SLEEP_INTERVAL = 3000;
const unsigned long SPI_TIMEOUT = 3000;
const unsigned long STARTUP_INTERVAL = 2000;
const unsigned long STARTUP_DELAY = 5000;
const unsigned long CANCEL_TIMEOUT = 5000;
const unsigned long BUTTON_TIMEOUT = 200;

// Thresholds
const float VIB_THRESHOLD = 1.0; // 1.0 = 1000 counts vibration
const int ROLL_THRESHOLD = 1000;
const int SW_THRESHOLD = 40;

// Buffers
char buffer[100];
char lastBuff[100], logBuff[100];

void setup() {
  Serial.begin(9600);
  pinMode(2, INPUT);
  pinMode(A1, INPUT);
  pinMode(A2, INPUT);
  pinMode(A3, INPUT);
  pinMode(A4, INPUT);
  pinMode(5, OUTPUT);
  pinMode(6, OUTPUT);
  digitalWrite(BUZZER_PIN, LOW);

  Serial.println("Smart Helmet v1.0 - With Cancel Button");
  Serial.println();
  Serial.println("Go Ready");
  Serial.println("Push button within 30 seconds to cancel alerts");
}

void loop() {
  checkStatus();

  if (helmetWarn) {
    readGPS();
    checkRoll();

    if (!startUpEvent && goReady && goReady2 && millis() > (lastStartUp + STARTUP_DELAY)) {
      startUpEvent = true;
    }

    checkAccident();
    checkEmergency();
    checkSleep();
    checkGSRCheck();
    checkVibration();
    checkRollAlert();
  }

  delay(100);

  // Check for Cancel Button
  if (digitalRead(CANCEL_BUTTON_PIN) == LOW) {
    current = digitalRead(IRAM_SENSOR_PIN) == LOW;

    if (current != helmetWarn) {
      helmetWarn = current;

      if (helmetWarn) {
        helmetAlert = millis();
        startUpEvent = false;
        waitingForCancel = false;
        alertCancelled = false;
        Serial.println("Cancel Button Pressed");
        delay(100);
      } else {
        Serial.println("Cancel Button Released");
        digitalWrite(BUZZER_PIN, LOW);
        accidentDetected = emergencyDetected = sleepDetected = false;
        waitingForCancel = false;
        alertCancelled = false;
      }
    }
  }
}

```

Figure C.1: complete source code